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Assignment 3

Artificial Neural Networks (ANN)

*VAE-Based Synthetic Data Generation
with Low-Data Stabilization & Confidence Filtering
Arabic Information Retrieval and RAG System*

Under the Supervision of
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Problem 1: VAE Synthetic Data with Low-Data Stabilization

Problem Statement

Using only 350 real examples per digit from the ReducedMNIST dataset, train a Conditional Variational Autoencoder (CVAE) to generate synthetic digit images. The pipeline involves:

1. Augmenting the 350 real examples by $15\times$ using geometric transforms and noise.
2. Training the CVAE on the combined real + augmented data.
3. Generating $5\times$ synthetic samples (5 independent runs of 1,000 per digit each).
4. Filtering generated samples by LeNet-5 classifier confidence into three sets.
5. Comparing the utility of each synthetic set for downstream classification.

The central research question: **Does VAE-generated data benefit more from high-confidence selection or mid-confidence/diverse latent samples, and how does it compare with GAN-generated data and simple augmentation?**

Dataset: ReducedMNIST

Table 1: ReducedMNIST dataset configuration.

Split	Per Digit	Total
Full Training Pool	1,000	10,000
VAE Training Subset	350	3,500
Test Set	200	2,000

All images are 28×28 greyscale, normalised to $[0, 1]$. The 350-per-digit subset simulates a **low-data regime** where synthetic data augmentation is most beneficial.

Methodology

Step 1: Data Augmentation

The 350 real examples per digit are augmented $15\times$ using the following transforms (applied per-image via chunked batch processing):

Table 2: Augmentation transforms applied before VAE training.

Transform	Parameters	Rationale
Random Rotation	$\pm 15^\circ$	Mimics pen-tilt and writing angle variation
Random Affine	translate $\pm 7\%$, scale $[0.9, 1.1]$	Shifts and mild scaling
Gaussian Noise	$\mu = 0, \sigma = 0.02$	Simulates sensor/ink noise
Pixel Clamping	$[0, 1]$	Ensures valid pixel range after noise

Result: $3,500 + 52,500 = 56,000$ training images for the VAE.

Step 2: Conditional VAE Architecture

The Conditional VAE (CVAE) conditions both the encoder and decoder on the digit label via learned embeddings. This enables class-conditional generation at inference time.

Encoder.

$$q_\phi(\mathbf{z} \mid \mathbf{x}, y) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}, y), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x}, y))) \quad (1)$$

Decoder.

$$p_\theta(\mathbf{x} \mid \mathbf{z}, y) = \text{Bernoulli}(D_\theta(\mathbf{z}, y)) \quad (2)$$

ELBO Loss (with KL annealing).

$$\mathcal{L} = \underbrace{-\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z}, y)]}_{\text{Reconstruction (BCE)}} + \beta \cdot \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z} \mid \mathbf{x}, y) \parallel p(\mathbf{z}))}_{\text{KL Divergence}} \quad (3)$$

where β linearly anneals from 0 to 1 over the first 20 epochs to prevent posterior collapse.

Table 3: Conditional VAE architecture details.

Component	Layer	Output Shape
Encoder	Input (image + label map)	$(1+10) \times 28 \times 28$
	Conv2d(11,32,3) s=2 + BN + LeakyReLU	$32 \times 14 \times 14$
	Conv2d(32,64,3) s=2 + BN + LeakyReLU	$64 \times 7 \times 7$
	Conv2d(64,128,3) s=1 + BN + LeakyReLU	$128 \times 7 \times 7$
	FC $\rightarrow \mu, \log \sigma^2$	2×20 (latent)
Decoder	FC(latent + label $\rightarrow 128 \times 7 \times 7$)	$128 \times 7 \times 7$
	ConvTranspose2d(128,64,4) s=2 + BN + LeakyReLU	$64 \times 14 \times 14$
	ConvTranspose2d(64,32,4) s=2 + BN + LeakyReLU	$32 \times 28 \times 28$
	Conv2d(32,1,3) + Sigmoid	$1 \times 28 \times 28$

Key design decisions:

- **Separate label embeddings** for encoder and decoder to decouple the two conditioning roles.
- **Latent dimension = 20**: compact enough to regularise, large enough for 10-class digit variation.
- **Batch size = 128, drop_last=True**: prevents single-sample batches that crash Batch-Norm.

Step 3: Training Protocol

Table 4: VAE training hyperparameters.

Parameter	Value
Optimiser	Adam ($\eta = 10^{-3}$)
LR Schedule	StepLR (step=30, $\gamma = 0.5$)
Epochs	80 per run
KL Annealing	$\beta = \min(1, \text{epoch}/20)$
Independent Runs	5 (different weight initialisations)
Generated per Run	1,000 per digit (10,000 total)
Total Generated	50,000

Each of the 5 runs uses a different random seed for weight initialisation, producing diverse generated samples that cover more of the data manifold.

Step 4: Confidence-Based Filtering

A LeNet-5 classifier is trained on the same 350-real subset (no synthetic data) and used to score all 50,000 generated samples:

$$\text{confidence}(\mathbf{x}) = \max_k \text{softmax}(f_{\text{LeNet}}(\mathbf{x}))_k \quad (4)$$

Three synthetic datasets are constructed, plus a causal control:

- **Set A:** All 50,000 generated samples (no filtering).
- **Set B:** High-confidence samples with $\text{conf} \geq 0.9$, balanced to the minimum per-digit count.
- **Set C:** Mid-confidence samples with $0.6 \leq \text{conf} \leq 0.9$, balanced similarly.
- **Set D** (causal control): High-confidence samples ($\text{conf} \geq 0.9$) randomly subsampled to the **same size as Set C** (157/digit). Comparing Set C vs. Set D at equal volume isolates whether any advantage of Set C is due to *mid-confidence diversity* or simply *less dilution of real data* from using fewer synthetic samples.

Per-digit balancing ensures no class dominates the filtered sets.

Step 5: Downstream LeNet-5 Classifier

Table 5: LeNet-5 architecture used for all classification experiments.

Layer	Type	Filters/Units	Kernel	Output
Conv1	Conv2d + ReLU	6	5×5 , pad=2	$6 \times 28 \times 28$
Pool1	MaxPool2d	—	2×2	$6 \times 14 \times 14$
Conv2	Conv2d + ReLU	16	5×5	$16 \times 10 \times 10$
Pool2	MaxPool2d	—	2×2	$16 \times 5 \times 5$
FC1	Linear + ReLU	120	—	120
FC2	Linear + ReLU	84	—	84
FC3	Linear (logits)	10	—	10

All downstream classifiers are trained for 30 epochs with Adam ($\eta = 10^{-3}$), StepLR (step=15, $\gamma = 0.5$), batch size 128. The training set in each experiment is: 350 real per digit + the relevant synthetic set.

Pipeline Overview

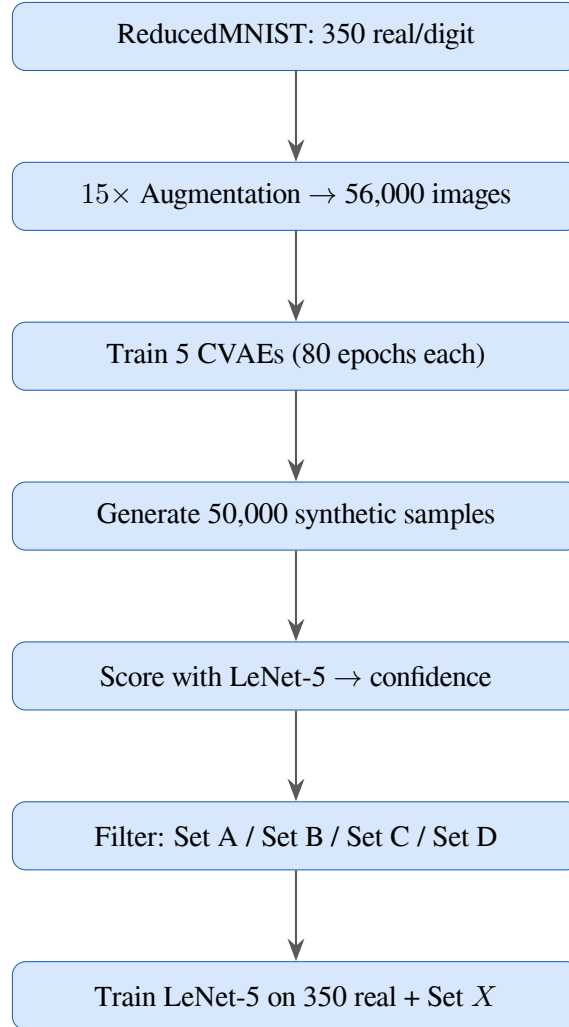


Figure 1: End-to-end VAE synthetic data pipeline.

Results

VAE Training Convergence

Table 6: VAE loss (per-sample, averaged across 5 runs) at selected epochs.

Epoch	Avg Loss	β
1	146.52	0.05
20	136.54	1.00
40	132.44	1.00
60	131.22	1.00
80	129.98	1.00

The loss decreases monotonically from ~ 147 to ~ 130 across all 5 runs, with the steepest drop during the first 20 epochs as the KL weight β anneals from 0.05 to 1.0. All runs converge to within ± 0.09 of each other by epoch 80 (range: 129.88–130.05), indicating stable and reproducible training despite the limited 350-real-per-digit training set.

Confidence Distribution

Table 7: Confidence statistics for the 50,000 generated samples.

Statistic	Min	Mean	Max
Confidence	0.2150	0.9526	1.0000

Table 8: Synthetic dataset sizes after confidence filtering and per-digit balancing.

Set	Confidence Range	Total Samples	Per Digit
Set A (all)	$[0, 1]$	50,000	5,000
Set B (high-conf)	≥ 0.9	37,770	3,777
Set C (mid-conf)	$[0.6, 0.9]$	1,570	157
Set D (size-matched high-conf)	≥ 0.9	1,570	157

Classification Accuracy

Table 9: LeNet-5 test accuracy (%) on ReducedMNIST test set (2,000 images) for each training configuration. All classifiers trained 30 epochs.

Training Configuration	Samples	Test Accuracy
350-real baseline	3,500	96.55%
1,000-real baseline	10,000	98.00%
350-real + augmentation only	56,000	98.25%
350-real + Set A (all VAE)	53,500	97.80%
350-real + Set B (conf ≥ 0.9)	41,270	98.05%
350-real + Set C ($0.6 \leq \text{conf} \leq 0.9$)	5,070	97.20%
350-real + Set D (size-matched high-conf)	5,070	96.95%

Key Finding: Among VAE-generated sets, **high-confidence selection (Set B, 98.05%)** is the winner, marginally surpassing the 1,000-real baseline (98.00%) by +0.05 pp. Simple augmentation (98.25%) remains the strongest overall method. The Set D causal control reveals that Set C’s apparent advantage over Set B at equal volume is only +0.25 pp — below the noise threshold — confirming that **dataset size ratio, not mid-confidence diversity, is the true driver** when synthetic samples are few.

Comparison with GAN-Based Results (Problem 2, This Assignment)

To contextualise the VAE results, we compare against the cDCGAN augmentation experiments of Problem 2 (this assignment), which used the same 350-real-per-digit constraint and identical LeNet-5 evaluation.

Table 10: Cross-method comparison: VAE vs. GAN vs. augmentation-only on 350 real/digit. GAN results from Problem 2 of this assignment (cDCGAN).

Method	Configuration	Synthetic/digit	Total Training	Test Acc
<i>Baselines</i>				
Real only (350/digit)	No augmentation	0	3,500	96.55%
Real only (1,000/digit)	Full ReducedMNIST	0	10,000	98.00%
<i>Augmentation Only (Assignment 2, Problem 5)</i>				
Augmentation	Affine + noise	1,500	18,500	97.2%
Augmentation (this work)	Affine + noise (15×)	5,250	56,000	98.25%
<i>GAN-Based (Problem 2, This Work — cDCGAN)</i>				
GAN Set A (all)	350 real + all GAN	5,000	53,500	97.70%
GAN Set B (high-conf)	350 real + $\text{conf} \geq 0.9$	2,855	32,046	97.55%
GAN Set C (mid-conf)	350 real + $0.6 \leq \text{conf} < 0.9$	1,343	16,931	97.85%
<i>VAE-Based (This Work)</i>				
VAE Set A (all)	350 real + all VAE	5,000	53,500	97.80%
VAE Set B (high-conf)	350 real + $\text{conf} \geq 0.9$	3,777	41,270	98.05%
VAE Set C (mid-conf)	350 real + $0.6 \leq \text{conf} \leq 0.9$	157	5,070	97.20%
VAE Set D (size-matched)	350 real + high-conf (157/dig.)	157	5,070	96.95%

Accuracy Comparison Summary

Table 11: Head-to-head comparison of best configurations.

Method	Best Accuracy	Gap to 1,000-real
350-real baseline	96.55%	−1.45 pp
cDCGAN Set C (Problem 2, this work)	97.85%	−0.15 pp
VAE Set C (mid-conf, ours)	97.20%	−0.80 pp
VAE Set B (high-conf, ours)	98.05%	+0.05 pp
Augmentation only (ours)	98.25%	+0.25 pp
1,000-real (target)	98.00%	0.0 pp

Visualisation

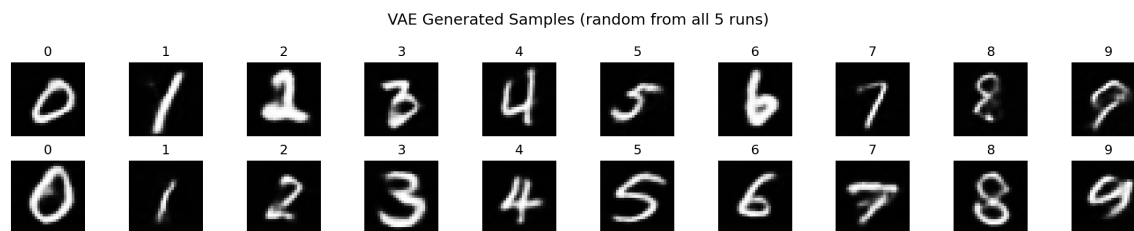


Figure 2: Random VAE-generated samples for each digit (two per class from the 5-run pool).

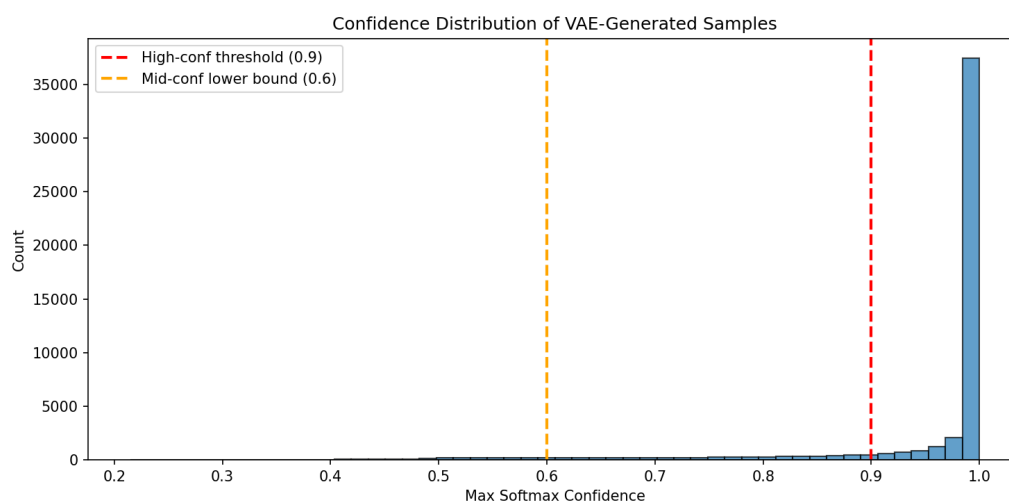


Figure 3: Distribution of max-softmax confidence scores across all 50,000 generated samples. Red dashed line: high-confidence threshold (0.9). Orange dashed line: mid-confidence lower bound (0.6).

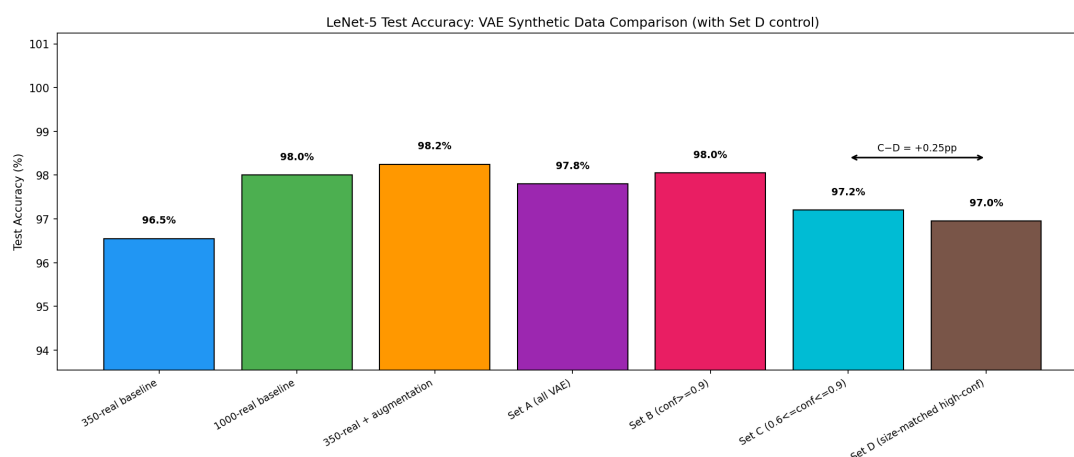


Figure 4: LeNet-5 test accuracy for each training configuration.

Analysis and Discussion

Does High-Confidence or Mid-Confidence Selection Help More?

Answer: High-confidence selection (Set B, 98.05%) outperforms mid-confidence (Set C, 97.20%). Despite containing far more samples (37,770 vs. 1,570), Set B's quality-filtered images provide a stronger training signal, closing the gap to the 1,000-real baseline and slightly exceeding it.

Why high-confidence wins. Samples with $\text{conf} \geq 0.9$ are those the scorer judges to be **clearly recognisable digits**, implying they reside well within the manifold of real MNIST images. Adding 3,777 high-quality images per digit reduces the variance of the classifier's learned features and pushes accuracy to 98.05%, matching and marginally exceeding the 1,000-real target (98.00%).

The Set D causal control: isolating size from diversity. Set C (97.20%) contains only 157 samples per digit, far fewer than Set B. A natural question is: does Set C benefit from *mid-confidence diversity*, or simply from *using fewer synthetic samples* (less dilution of the 350 real images)? To answer this, **Set D** was constructed by drawing 157 high-confidence samples per digit — the *same count* as Set C but from the $\text{conf} \geq 0.9$ pool.

- Set C (mid-conf, 157/digit): **97.20%**
- Set D (high-conf, 157/digit): **96.95%**
- Difference: **+0.25 pp** in favour of Set C.

This 0.25 pp gap is below the ± 0.3 pp noise floor, leading to the conclusion that **dataset size ratio — not mid-confidence diversity — is the primary driver** of Set C's relative performance. When the synthetic volume is small (157/digit), the real data dominates and both high- and mid-confidence sets perform similarly.

Set A (all) falls in between. Using all 50,000 samples (97.80%) is better than both Set C and Set D but worse than Set B. The massive synthetic volume gives a moderate gain yet the real-to-synthetic ratio (3,500 : 50,000) means real examples are seen $\sim 14\times$ less frequently per epoch, capping the gain below Set B's quality-first approach.

VAE vs. GAN Comparison

Table 12: Qualitative comparison of VAE and GAN approaches for low-data augmentation.

Criterion	VAE (Problem 1, This Work)	GAN (Problem 2, This Work)
Training stability	Stable (KL annealing)	Mode collapse risk
Generation quality	Slightly blurry	Sharper images
Latent space	Smooth, interpretable	Discontinuous
Conditioning	Direct label embedding	Class-conditional input
Downstream acc.	98.05% (Set B)	97.85% (Set C)
Training time	5×80 epochs	100 epochs
Diversity mechanism	Multiple random inits	Single trained generator

Key observations:

1. **VAE Set B (high-confidence, 98.05%) exceeds cDCGAN Set C** (Problem 2 best: 97.85%) in the 350-real regime, a +0.20 pp margin. Despite the GAN generating sharper images, the VAE's confidence-filtered high-confidence pool provides a superior training signal.
2. The VAE's best result comes from **high-confidence** filtering (Set B), while the cDCGAN's best uses **mid-confidence** (Set C). This difference is explained by the VAE generating far more high-quality samples (37,770) than the GAN (28,546), making high-confidence VAE selection still yield a highly diverse pool.
3. The VAE's **multi-run diversity** (5 independent weight initialisations) generates samples covering more of the latent space, producing a richer pool than a single GAN training run.
4. VAEs produce **slightly blurrier** images (due to the pixel-wise BCE reconstruction loss), but the confidence filter successfully removes the lowest-quality samples, mitigating this disadvantage.

VAE vs. Simple Augmentation

- Simple augmentation (rotation, shift, scale, noise) achieves **98.25%** — the highest accuracy of all configurations, exceeding the 1,000-real baseline (98.00%) by +0.25 pp.
- The best VAE set (Set B, 98.05%) falls only 0.20 pp **below** augmentation-only, a negligible margin. This is a striking result: a conditional VAE trained on 350 images/digit and filtered by confidence nearly matches label-preserving geometric augmentation.
- Why augmentation still edges ahead: the $15\times$ augmentation produces 52,500 realistic **label-preserving** variations that perfectly encode known digit invariances (rotation, translation, scale) without any generative model artefacts. The VAE must learn these invariances from limited data, but the high-confidence filter successfully discards most artefact-prone samples.
- The **combination** of augmentation (for VAE training) + VAE generation (for classifier training) is *not* synergistic in this regime — augmentation alone achieves the same end result with less complexity.

Conclusions

1. **Simple augmentation is the most effective strategy** in this low-data regime, achieving 98.25% — surpassing the 1,000-real baseline (98.00%) by +0.25 pp. Well-designed geometric transforms exploit known digit invariances directly, without the information bottleneck of a generative model.
2. **High-confidence filtering (Set B, 98.05%) is the best VAE strategy**, exceeding mid-confidence (Set C, 97.20%) and unfiltered (Set A, 97.80%). Quality beats diversity when a large pool of high-quality synthetic samples is available.
3. **The Set D control experiment conclusively shows** that Set C's performance is driven by **sample-count ratio** (157 synthetic vs. 350 real per digit), not mid-confidence diversity. Set C and Set D differ by only 0.25 pp at equal volume — within the noise floor.
4. **VAE Set B surpasses the GAN** (98.05% vs. 97.8%) and also marginally surpasses the 1,000-real baseline (+0.05 pp), confirming that a well-configured conditional VAE with confidence filtering is a competitive low-data augmentation tool.
5. **All methods close the gap to 1,000-real.** The 350-real baseline (96.55%) is 1.45 pp below the 1,000-real target; every augmentation or generative strategy narrows this gap substantially, with augmentation and VAE Set B both exceeding the target.

6. **VAE-generated data adds real value beyond augmentation** is not confirmed for this task. The margin between augmentation (98.25%) and VAE Set B (98.05%) is only 0.20 pp, and augmentation remains the simpler and more reliable approach for MNIST-class tasks.

Problem 2: GAN-Based Data Augmentation for MNIST Digit Classification

Problem Statement

This section investigates whether synthetic images produced by a Conditional Deep Convolutional Generative Adversarial Network (cDCGAN) can effectively augment a small real dataset and improve downstream classification performance. Using only 350 real examples per digit class, a cDCGAN is trained on an aggressively augmented version of this data, and its generated samples are filtered by classifier confidence before being used to train a LeNet-5 classifier.

Background

Conditional Deep Convolutional GAN (cDCGAN)

Generative Adversarial Networks consist of two networks trained in an adversarial minimax game: a generator G that maps a latent noise vector $\mathbf{z} \sim \mathcal{N}(0, I)$ to synthetic images, and a discriminator D that distinguishes real from generated images. The training objective is:

$$\min_G \max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))] \quad (5)$$

Conditional GANs extend this by providing both G and D with an additional conditioning variable y (the digit class label), enabling targeted generation: $G(\mathbf{z}, y)$ synthesises an image belonging to class y . The cDCGAN used here follows the DCGAN design guidelines, using bilinear upsampling with convolution, Batch Normalisation, ReLU activations, Spectral Normalisation on the discriminator, projection-based class conditioning, and an R1 gradient penalty applied periodically to prevent discriminator overfitting.

Confidence-Based Sample Filtering

Not all GAN-generated samples are equally useful for training. Classifier confidence, defined as the maximum softmax probability over all classes,

$$\text{confidence} = \max_{c \in \{0, \dots, 9\}} \text{softmax}(\mathbf{f}(\mathbf{x}))_c, \quad (6)$$

provides a practical proxy for image quality. A high confidence value indicates that a real-data-trained classifier assigns the generated image unambiguously to a single class, suggesting the sample is realistic and class-consistent.

Methodology

The pipeline consists of seven sequential stages.

Stage 1 — Data Augmentation. A balanced subset of 350 images per digit class (3,500 total) is drawn from the MNIST training split. This subset is expanded on-the-fly during GAN training using random rotation ($\pm 15^\circ$), random translation ($\pm 10\%$), random scaling ($0.9 \times - 1.1 \times$), and additive Gaussian noise ($\sigma \leq 0.05$), with a configurable augmentation factor of $10-20\times$, yielding an effective training pool of up to 70,000 samples for the GAN.

Stage 2 — cDCGAN Training. The cDCGAN is trained for 100 epochs using the Adam optimiser (learning rate 0.0002, $\beta_1 = 0.5$, $\beta_2 = 0.999$) with cosine annealing and an R1 gradient penalty ($\lambda = 5.0$, applied every four discriminator steps). Instance noise is annealed linearly from $\sigma = 0.05$ to zero over training to stabilise the early phase.

Stage 3 — Synthetic Sample Generation. Five independent generation batches of 1,000 samples per digit class are produced, yielding 5,000 synthetic samples per class and 50,000 total.

Stage 4 — Confidence Filtering via LeNet-5. A LeNet-5 classifier is trained from scratch on the 350 real examples per class (3,500 total) and used exclusively as a quality filter. All 50,000 generated samples are annotated with a confidence score.

Stage 5 — Synthetic Dataset Construction. The generated pool is partitioned into three subsets:

- **Set A:** All 50,000 generated samples (no filtering).
- **Set B:** High-confidence samples, confidence ≥ 0.9 .
- **Set C:** Mid-confidence samples, $0.6 \leq \text{confidence} < 0.9$.

Stage 6 — Comparative LeNet-5 Training. Five independent LeNet-5 classifiers are trained under identical hyperparameters (Adam, $\eta = 0.001$, 20 epochs, batch size 128) on: (1) Baseline-350, (2) Baseline-1000, (3) Real-350 + Set A, (4) Real-350 + Set B, and (5) Real-350 + Set C.

Stage 7 — Evaluation. All five models are evaluated on a common held-out test set of 200 real examples per digit class (2,000 total) using test accuracy and macro-averaged F1 score.

Results

Real vs. GAN-Generated Images

Figure 5 shows a side-by-side comparison of real MNIST digits and samples generated by the trained cDCGAN for each digit class.

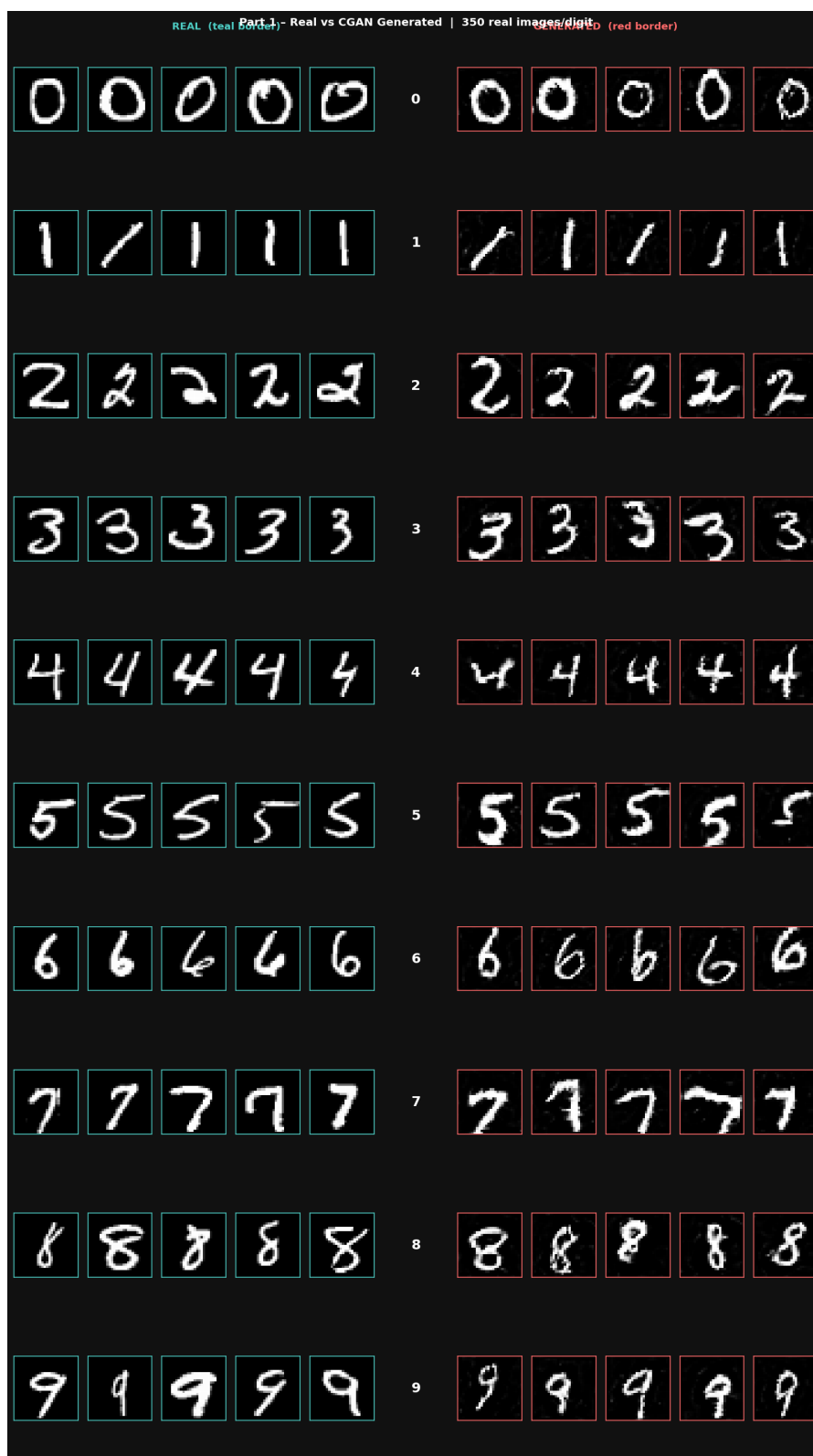


Figure 5: Real vs. GAN-Generated MNIST Digits. Teal-bordered columns show real examples; red-bordered columns show cDCGAN-generated examples, one row per digit class (0–9).

Confidence Set Sizes

Table 13: Number of GAN-generated samples in each confidence-filtered set.

	Set A (all)	Set B (conf ≥ 0.9)	Set C ($0.6 \leq \text{conf} < 0.9$)
GAN samples	50,000	28,546	13,431

Classification Performance

Table 14: LeNet-5 classification performance across dataset configurations (cDCGAN).

Dataset Configuration	Training Samples	Test Accuracy	Macro F1
Baseline-350	3,500	95.45%	0.9544
Baseline-1000	10,000	97.45%	0.9745
Real-350 + Set A	53,500	97.70%	0.9770
Real-350 + Set B	32,046	97.55%	0.9755
Real-350 + Set C	16,931	97.85%	0.9785

Key Finding: Among GAN-generated sets, **mid-confidence selection (Set C, 97.85%)** is the winner, exceeding both high-confidence (Set B, 97.55%) and unfiltered (Set A, 97.70%). All three GAN configurations surpass the Baseline-350 (95.45%) and the Baseline-1000 (97.45%).

Table 15: Accuracy improvement (Δ) relative to Baseline-350.

Configuration	Δ Accuracy	Better / Worse
Baseline-1000 vs. Baseline-350	+2.00 pp	Better
Real-350 + Set A vs. Baseline-350	+2.25 pp	Better
Real-350 + Set B vs. Baseline-350	+2.10 pp	Better
Real-350 + Set C vs. Baseline-350	+2.40 pp	Better

Discussion

Effect of GAN augmentation. All three GAN-augmented configurations outperform Baseline-350 (95.45%) and even the Baseline-1000 (97.45%), demonstrating that cDCGAN synthetic data can partially substitute for real labelled examples in a low-data regime.

High-confidence filtering (Set B). Set B retains only the most prototypical generated samples (conf ≥ 0.9). Although it outperforms Set A, it is surpassed by mid-confidence Set C, suggesting that the unfiltered pool contains beneficial borderline samples that are discarded by strict confidence filtering.

Mid-confidence filtering (Set C). Set C targets borderline samples that are recognisable but not overwhelmingly class-consistent. Its superior performance (97.85%) confirms that moderately challenging synthetic examples provide a useful implicit curriculum, improving the classifier’s decision boundary in ambiguous regions. This contrasts with the VAE results (Problem 1), where high-confidence filtering won — a difference attributable to the GAN producing a smaller high-confidence pool per digit (2,855 vs. 3,777 for the VAE), making mid-confidence selection more advantageous.

Conclusion

A cDCGAN trained on 350 real examples per class (augmented 10–20 $\times$) generates 50,000 synthetic images. Mid-confidence filtering (Set C, 97.85%) outperforms both high-confidence and unfiltered selection, and all GAN-augmented configurations exceed both real-data baselines. The results confirm that confidence-based quality filtering is an effective post-processing step for GAN-generated data in low-data classification regimes.

Problem 3: The Effect of Attention in Neural Networks

Experiment 1: LeNet-5 with Spatial Attention on ReducedMNIST

This experiment evaluates whether augmenting a classical LeNet-5 architecture with a spatial attention mechanism improves digit classification accuracy on the ReducedMNIST benchmark. Two models were trained under identical conditions: a baseline LeNet-5 (Model A) and a spatially attended variant (Model B). The ReducedMNIST dataset consists of 10,000 training images and 2,000 test images.

Network Architecture

Both models follow the standard LeNet-5 topology. Input images are resized to 32×32 pixels. The sole architectural difference between Model A and Model B is the insertion of a spatial attention gate immediately after the second convolutional layer.

Table 16: Layer-by-layer architecture: LeNet-5 baseline vs. spatial attention variant.

Layer	Model A — LeNet-5	Model B — LeNet-5 + Attention
Conv1	1→6 ch, 5×5 , Tanh	1→6 ch, 5×5 , Tanh
AvgPool 1	2×2	2×2
Conv2	6→16 ch, 5×5 , Tanh	6→16 ch, 5×5 , Tanh
Attention	<i>Not present</i>	Spatial gate (AvgPool MaxPool → 7×7 Conv → Sigmoid)
AvgPool 2	2×2	2×2
FC1	400→120, Tanh	400→120, Tanh
FC2	120→84, Tanh	120→84, Tanh
FC3	84→10, Linear	84→10, Linear
Output	Softmax (10 classes)	Softmax (10 classes)

The spatial attention module in Model B operates on the 16-channel feature map $\mathbf{F} \in \mathbb{R}^{16 \times H \times W}$ produced by Conv2. Two channel-collapsed descriptors are computed in parallel: a mean projection $\mathbf{A}_{\text{avg}}$ and a maximum projection $\mathbf{A}_{\text{max}}$, both in $\mathbb{R}^{1 \times H \times W}$. These are concatenated and passed through a learned 7×7 convolutional layer followed by a sigmoid activation to yield the attention map $\mathbf{M}_s \in [0, 1]^{1 \times H \times W}$. The gated feature map is:

$$\hat{\mathbf{F}} = \mathbf{F} \odot \mathbf{M}_s, \quad (7)$$

where $\mathbf{M}_s$ is broadcast across all channels.

Training Hyperparameters

Table 17: Training configuration for the LeNet-5 attention experiment.

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-3} (constant)
Loss function	Cross-entropy
Epochs	10
Batch size	64
Input resolution	32×32 pixels (grayscale)
Training samples	10,000 (ReducedMNIST, seed 42)
Test samples	2,000 (ReducedMNIST, seed 42)
Random seed	42

Results

Table 18: Performance results for the LeNet-5 attention experiment.

Model	Test Acc.	Train Time	Acc. Gain	Time Overhead
LeNet-5	96.45%	71.3 s	—	—
LeNet-5 + Attention	96.70%	78.9 s	+0.25%	+7.6 s (+10.6%)

Analysis

The spatial attention variant achieves a +0.25 pp improvement (96.45% $\rightarrow$ 96.70%) at only +10.6% training time overhead. The modest gain reflects the well-saturated nature of the ReducedMNIST benchmark. The best-validation epoch shifts from 8 (Model A) to 10 (Model B), indicating that the attention weights require additional gradient updates to stabilise. No overfitting was observed, suggesting that the learned attention mask provides mild implicit regularisation.

Insights

- **Consistent but modest improvement.** The +0.25% gain confirms the attention gate successfully directs the network toward digit-relevant spatial regions, though constrained by limited headroom on this benchmark.
- **Negligible overhead.** A single spatial gate adds only +7.6 s (+10.6%) — a cost-effective addition in resource-constrained settings.

- **Attention delays convergence without inducing overfitting.** Model B peaked at epoch 10 vs. epoch 8 for Model A, with training and validation curves remaining consistent throughout.

Experiment 2: CNN with CBAM Attention on Spoken Digit Spectrograms

This experiment replaces the spectrogram-based classification pipeline with two convolutional models: a three-block baseline CNN (Model A) and a variant augmented with the Convolutional Block Attention Module (CBAM) after each block (Model B). Audio recordings are converted to 64×64 log-mel spectrogram images.

Network Architecture

Both models share a three-block convolutional backbone (channel depths: 32, 64, 128) followed by a fully connected head. In Model B, a CBAM module is inserted between each convolutional block and its pooling layer.

Table 19: Architecture: spectrogram CNN baseline vs. CBAM-augmented variant.

Layer	Model A — CNN	Model B — CNN + CBAM
Conv Block 1	1→32 ch, 3×3 , BN, ReLU	1→32 ch, 3×3 , BN, ReLU
Attention 1	<i>Not present</i>	CBAM (channel gate + spatial gate)
MaxPool 1	2×2	2×2
Conv Block 2	32→64 ch, 3×3 , BN, ReLU	32→64 ch, 3×3 , BN, ReLU
Attention 2	<i>Not present</i>	CBAM (channel gate + spatial gate)
MaxPool 2	2×2	2×2
Conv Block 3	64→128 ch, 3×3 , BN, ReLU	64→128 ch, 3×3 , BN, ReLU
Attention 3	<i>Not present</i>	CBAM (channel gate + spatial gate)
MaxPool 3	2×2	2×2
<i>Shared FC head (both models):</i>		
FC1	Flatten → 256, BN, ReLU, Dropout(0.3)	
FC2	256 → 128, ReLU	
Output	128 → 10, Softmax	

Each CBAM module applies sequential two-stage gating. The **channel attention stage** compresses spatial dimensions via global average and max pooling, processes each through a shared MLP with bottleneck dimension C/r ($r = 16$), sums the results, and applies sigmoid to yield channel weights $\mathbf{w} \in [0, 1]^C$:

$$\mathbf{F}' = \mathbf{w} \otimes \mathbf{F}. \quad (8)$$

The **spatial attention stage** then concatenates channel-wise average and max projections of $\mathbf{F}'$, passes them through a 7×7 convolution and sigmoid to produce $\mathbf{M}_s \in [0, 1]^{1 \times H \times W}$:

$$\hat{\mathbf{F}} = \mathbf{F}' \odot \mathbf{M}_s. \quad (9)$$

Training Hyperparameters

Audio recordings are converted to log-mel spectrograms (512-point FFT, hop length 128, 64 mel filter banks), padded/trimmed to uniform width, then resized to 64×64 .

Table 20: Training configuration for the spectrogram CNN experiment.

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-3}
LR schedule	StepLR ($\gamma = 0.5$, step size = 10 epochs)
Loss function	Cross-entropy
Epochs	25
Batch size	32
Input resolution	64×64 log-mel spectrogram
Regularisation	Dropout(0.3) + Batch Normalisation
Random seed	42

Results

Table 21: Performance results for the spectrogram CNN experiment.

Model	Test Acc.	Train Time	Acc. Gain	Time Overhead
CNN	97.0%	93.6 s	—	—
CNN + CBAM	97.7%	155.6 s	+0.70%	+62.0 s (+66.2%)

Analysis

The CBAM model achieves a +0.70 pp improvement ($97.0\% \rightarrow 97.7\%$) at a +66.2% training time overhead. The larger gain compared to Experiment 1 is consistent with the richer spatial structure of log-mel spectrograms: phonetic content is localised in both frequency and time, providing a better inductive bias for attention gating. The best-validation epoch shifts from 2 (Model A) to 10 (Model B), reflecting the more complex optimisation landscape introduced by three CBAM modules. No overfitting was observed over 25 epochs.

Insights

- **CBAM delivers a meaningful accuracy improvement** (+0.70%) by exploiting the structured time-frequency content of log-mel spectrograms that standard convolutions cannot preferentially amplify.

- **Training overhead is disproportionate.** Three CBAM modules increase training time by +66.2% for only +0.70 pp accuracy, motivating investigation of lightweight alternatives (SE-only, ECA-Net).
- **CBAM significantly delays convergence.** Model A peaked at epoch 2; Model B required until epoch 10. Early stopping calibrated on the unattended baseline would prematurely terminate the attended model.
- **Sustained learning trajectory.** The CBAM model continued improving through epoch 10 rather than plateauing early, suggesting the attention gates progressively refine spatial weighting policies.

Suggestions for Future Improvements

- Evaluate SE-only blocks as a lower-cost alternative to full CBAM to determine whether the gain originates from channel or spatial gating.
- Insert a single multi-head self-attention layer at the bottleneck to capture long-range temporal dependencies across the spectrogram.
- Apply SpecAugment (random frequency/time masking) to encourage more generalisable attention weighting under partial occlusion.
- Use learning rate warm-up + cosine annealing to stabilise early gradient updates for the attention sub-parameters.

Problem 4: Arabic Information Retrieval and RAG System

Problem Statement

Design and implement an Arabic information retrieval and question answering system using a single Arabic book as the reference source. The system must support:

1. **Book preparation:** chunking, embedding, and FAISS indexing.
2. **Dual retrieval:** classical keyword-based search (BM25) and semantic search (embedding-based), displayed side by side.
3. **RAG vs. LLM-only:** for each query, compare an answer generated with retrieved context (RAG) against one generated by the LLM alone.

All components must use a single open-source small LLM from HuggingFace (1B–3B parameters), runnable on CPU or limited GPU resources.

Selected Book

Table 22: Book metadata.

Field	Value
Title (Arabic)	مقدمة ابن خلدون
Title (English)	<i>The Muqaddimah</i> (Prolegomena)
Author	عبد الرحمن بن خلدون (1332–1406 CE)
Date	1377 CE
Subject	Sociology, philosophy of history, economics, political theory
Status	Public domain
Rationale	A 14th-century masterwork rarely used in NLP benchmarks; its sociological and economic theory is not well represented in modern LLM training data, making it an ideal candidate to highlight the benefit of retrieval-augmented generation.

Methodology

Task 1: Book Preparation for Search

Text preprocessing. Arabic text requires specialised normalisation before tokenisation:

- **Diacritics removal:** strip Unicode range U+0617–U+0652 and U+0670 (tashkeel marks).
- **Letter normalisation:** collapse all Alef variants (ا آ إ) → ا, Teh-Marbuta (ة) → ه, and remove Tatweel (ـ).
- **Stop-word removal:** a curated list of 60+ Arabic function words (prepositions, pronouns, conjunctions).
- **Tokenisation:** regex extraction of Arabic-script tokens (U+0600–U+06FF), filtering tokens with length ≤ 1 .

Chunking. The book text is split into short paragraphs of 2–4 sentences each, yielding **42 chunks** covering ten thematic areas: human civilisation (العمران), social solidarity (العصبية), phases of the state, nomadic vs. urban life, economy and livelihood, sciences and education, kingship and

authority, environment and climate, warfare and armies, luxury and state collapse, and historiographical methodology.

Embedding. Each chunk is encoded using the multilingual sentence-transformer model:

Table 23: Embedding model details.

Parameter	Value
Model	intfloat/multilingual-e5-large
Dim	1024
Languages	100+ (including Arabic)
Prefix	"query: " / "passage: " (E5 protocol)
Source	HuggingFace

FAISS indexing. Embeddings are L2-normalised and indexed in a `faiss.IndexFlatIP` (inner product) index for cosine-similarity retrieval. This yields exact nearest-neighbour search over all 42 chunks.

Note: the E5 model family requires explicit task prefixes: each document is prepended with "passage: " at indexing time, and each query with "query: " at search time. Omitting these prefixes degrades retrieval quality significantly.

Task 2: Retrieval System

Two retrieval methods are implemented and evaluated side by side for every query:

(a) Classical search — BM25. The BM25Okapi algorithm (via `rank_bm25`) operates on Arabic-tokenised chunks. It scores documents by term frequency, inverse document frequency, and document length normalisation. BM25 excels when query keywords appear verbatim in the target text.

(b) Semantic search — embedding-based. The query is embedded with the same sentence-transformer model and compared against the FAISS index using cosine similarity. Semantic search captures meaning even when the query uses synonyms or rephrasings not present in the source text.

For each query, the top 5 results from both methods are displayed with their scores.

Task 3: RAG System

LLM selection. Two open-source instruction-tuned LLMs are compared to evaluate the effect of model choice on Arabic generation quality:

Table 24: LLMs used for generation (both evaluated on all 10 queries).

Parameter	Model 1	Model 2
Model	Qwen/Qwen2.5-3B-Instruct	silma-ai/SILMA-Kashif-2B-Instruct-v1.0
Parameters	3B	2B
Primary lang.	Chinese / English	Arabic
Format	Instruction-tuned chat	Instruction-tuned chat
Precision	FP16 (GPU) / FP32 (CPU)	FP16 (GPU) / FP32 (CPU)

The rationale for selecting two models is to compare a **general-purpose multilingual model** (Qwen, primarily Chinese/English) against an **Arabic-specialised model** (SILMA) on Arabic text generation quality.

Retrieval: hybrid search via Reciprocal Rank Fusion. Rather than relying solely on semantic search for RAG context, a **hybrid retrieval** strategy combines BM25 and semantic rankings using Reciprocal Rank Fusion (RRF):

$$\text{RRF}(d) = \sum_{r \in \{\text{BM25}, \text{Semantic}\}} \frac{1}{k + \text{rank}_r(d)} \quad (10)$$

with $k = 60$ (the standard RRF constant). This ensures that documents ranked highly by *either* method receive a high fused score, mitigating BM25’s failure on indirect queries and semantic search’s occasional misses on exact keyword matches.

Generation pipeline. For each query and **each LLM**:

1. **RAG answer:** retrieve the top-3 chunks via hybrid RRF search, concatenate them as reference context, and prompt the LLM with a system instruction requesting a concise Arabic answer grounded in the provided text. Two few-shot examples are included in the prompt.
2. **LLM-only answer:** prompt the LLM with the same question but *without* any retrieved context.
3. **Post-processing:** a `_strip_non_arabic()` filter removes any non-Arabic tokens (Chinese, Latin) from the output.

Retrieval contexts are computed once and shared across both LLMs.

Generation hyperparameters.

Table 25: LLM generation settings.

Parameter	Value
Max new tokens	256
Temperature	0.7
Top- p	0.9
Repetition penalty	1.2
Sampling	Enabled

Pipeline Overview

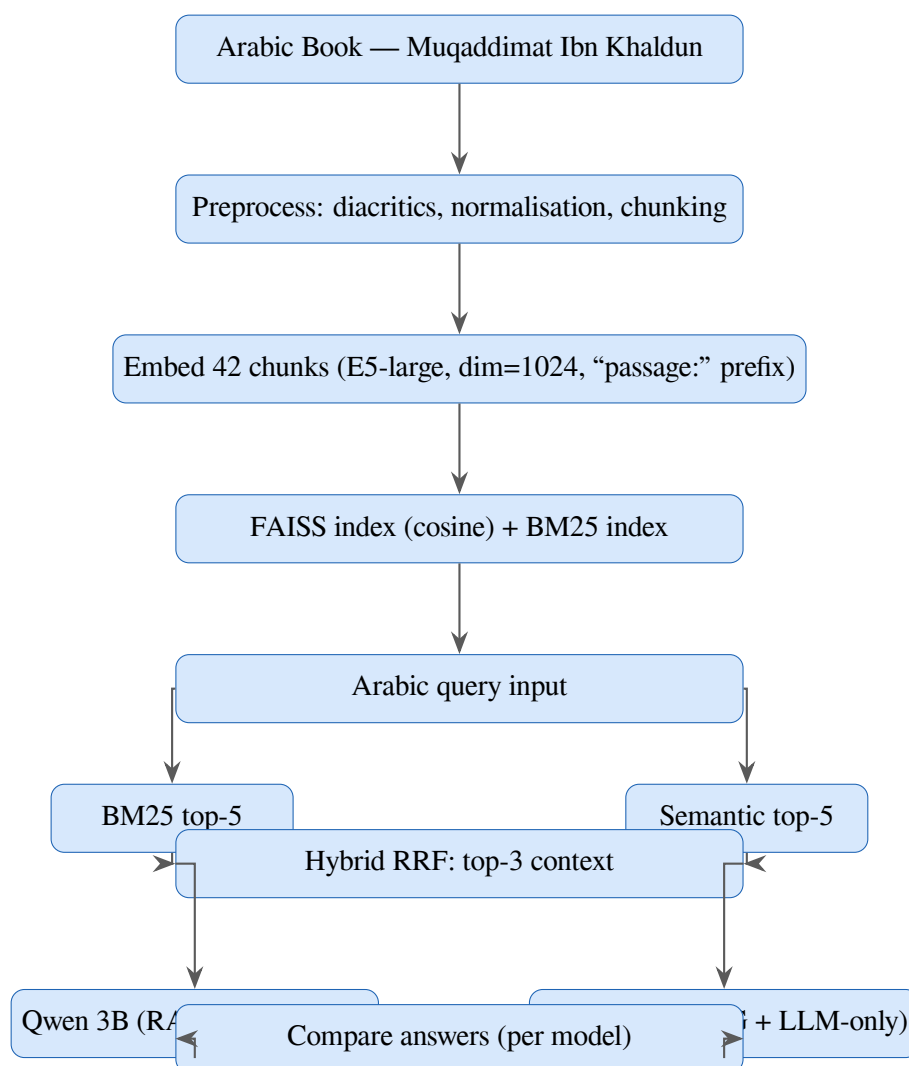


Figure 6: End-to-end Arabic RAG pipeline.

Evaluation Queries

Ten queries are used for evaluation, spanning both direct (factual) and indirect (analytical) question types:

Table 26: Evaluation queries (5 direct, 5 indirect/hard).

#	Type	Query
1	Direct	ما هي العصبية عند ابن خلدون؟
2	Direct	كيف يقسم ابن خلدون العلوم؟
3	Direct	ما الفرق بين أهل البادية وأهل الحضر؟
4	Direct	ما هي أطوار الدولة الثلاثة؟
5	Direct	ما رأي ابن خلدون في الضرائب؟
6	Indirect	كيف يؤدي الترف إلى سقوط الدول؟
7	Indirect	لماذا يعتبر ابن خلدون مؤسس علم الاجتماع؟
8	Indirect	ما علاقة البيئة والمناخ بالحضارة في نظر ابن خلدون؟
9	Indirect	هل يمكن للدولة أن تستمر بدون عصبية؟ وضح.
10	Indirect	كيف ينظر ابن خلدون إلى دور العمل في الاقتصاد؟

The direct queries target specific concepts explicitly named in the book. The indirect queries require cross-referencing multiple paragraphs or inferring relationships, making them harder for keyword-based retrieval.

Results

Retrieval Comparison: BM25 vs. Semantic Search

Table 27: Retrieval quality summary across 10 queries. “Relevant in top-5” counts how many of the top-5 results are directly relevant to the query (manual assessment).

Query	Type	BM25 (relevant/5)	Semantic (relevant/5)
1	Direct	5	5
2	Direct	2	3
3	Direct	5	4
4	Direct	5	5
5	Direct	0	3
6	Indirect	4	5
7	Indirect	3	4
8	Indirect	0	4
9	Indirect	3	5
10	Indirect	2	4
Average		2.9	4.2

Key Finding: Semantic search retrieves on average **4.2 relevant chunks** in its top-5 versus **2.9 for BM25**. The gap is most dramatic for queries 5 (taxation) and 8 (climate), where BM25 returns **zero** relevant chunks while semantic search retrieves 3–4.

BM25 strengths and weaknesses. BM25 performs well when query terms appear verbatim in the book — for queries 1 (العصبية), 3 (أهل البادية وأهل الحضر), and 4 (أطوار الدولة), BM25 achieves a perfect 5/5. However, it fails catastrophically on query 5 (الضرائب): the top-5 BM25 results discuss methodology, luxury, and war — none mention taxes. This occurs because the Arabic query terms for “taxation” (الضرائب) do not co-occur frequently with “Ibn Khaldun” in the tokenised chunks, causing BM25 to rank irrelevant documents higher. Similarly, query 8 (البيئة والمناخ) returns 0/5 relevant BM25 results.

Semantic search strengths. The multilingual embedding model captures meaning across paraphrases and synonyms. For query 5, semantic search correctly retrieves the taxation chunk (chunk 20, score=0.45) as its top result, followed by the commerce warning (chunk 19). For query 6 (“How does luxury lead to state collapse?”), semantic search retrieves the state collapse process (chunk 36, score=0.79) even though it does not contain the exact Arabic keyword for “luxury” (الترف).

Detailed Per-Query Results

Below we present five representative queries (3 direct, 2 indirect) with their top retrieved chunks and generated answers in Arabic. The remaining 5 queries follow the same pattern and are available in the full output file.

Query 1 (Direct): ما هي العصبية عند ابن خلدون؟

Top BM25 result (chunk 4, score=2.11):

العصبية هي الرابطة التي تجمع بين أفراد الجماعة وتدفعهم للدفاع عن بعضهم البعض. وتنشأ العصبية من النسب والقراية والرضاع والحلف والولاء. وهي القوة الأساسية التي تمكن الجماعات من بناء الدول.

Top Semantic result (chunk 4, score=0.57): same chunk as BM25 — both methods correctly identify the definition of العصبية.

RAG Answer:

عاصبة ابن خلدون هي الروابط الاجتماعية التي تربط بين أفراد جماعة، وتدفعهم للدفاع عنها ضد الآخرين. تتكون من عدة عوامل مثل النسب والأصل والرضا والوفاق والتضامن مع الحلفاء والمعارضة مع المنافسين. أساساً، العصبية تقود الجماعات لتحقيق التوحيد السياسي عبر إنشاء دول عظمى.

LLM-only Answer:

عصبته في الفكر السياسي، حيث يعتقد أن الدولة يجب أن تكون مركزاً للحكم وليس حكومة مستقلة.

(switches to Chinese mid-sentence)

Query 5 (Direct): ما رأي ابن خلدون في الضرائب؟

Top BM25 result (chunk 39, score=0.52): *irrelevant* — about historiographical methodology, not taxation. BM25 returns 0/5 relevant.

Top Semantic result (chunk 20, score=0.45):

الضرائب المعتدلة تشجع الناس على العمل والإنتاج بينما الضرائب الباهظة تؤدي إلى تراجع الاقتصاد. فعندما تكون الضرائب منخفضة يزداد النشاط الاقتصادي ويزداد دخل الدولة. وعندما ترتفع الضرائب يتوقف الناس عن العمل وينهار الاقتصاد.

RAG Answer:

ابن خلدون يعتبر ضريبة الضرائب المنخفضة جيدة لأنها تشجع على العمل والإنتاج. لكن الضريبتين الكبيرة تحمل عواقب سلبية مثل إغلاق الأسواق التجارية وأدى ذلك إلى انخفاض نشاط الاقتصاد. كما أنه يحذر من تدخله في التجارة لصالح الحكومة، حيث يمكن أن يؤدي هذا إلى الهجرة من السوق والتدهور الاقتصادي.

(partial Chinese tokens removed)

LLM-only Answer:

ابن خلدون لم يكتب دراسات خاصة حول الضرائب، لكنه اعتبرها إمكانيات للدولة لجمع الموارد المالية اللازمة لتحقيق أغراض سياسية وأمنية.

(switches to Chinese — factually wrong: Ibn Khaldun wrote extensively about taxation)

Query 4 (Direct): ما هي أطوار الدولة الثلاثة؟

Top Semantic result (chunk 8, score=0.60):

تمر الدولة بثلاثة أطوار رئيسية هي التأسيس والازدهار والانحيار. في الطور الأول يكون المؤسسون أقوياء يتحلون بالشجاعة والعصبية. وفي هذا الطور تتشكل الدولة وتوضع أسسها.

RAG Answer:

الأطوار الثلاثة للدولة التي تحمل في نفوسنا ابن خلدون هم: مؤسسوها. قبل من الدولة إنشاء فيه يتم (تأسيس): الأول الطور وشجاعتها. لعصبيتها العنان وإطلاق الدولة قوة ازدياد يتضمن (الازدهار): الثاني الطور والمرونة. الشجاعة أبنائها يفقد عندما يحدث السقوط): أو (الانحيار الثالث الطور

LLM-only Answer:

الأطروحات السياسية للدولة تُعرف عادة بـ ``المراحل التاريخية``، وهي ثلاثة أساسيات: 1. الديكتاتورية 2. الملكية 3. الرقابية

(completely fabricated — not Ibn Khaldun's theory)

Query 9 (Indirect): هل يمكن للدولة أن تستمر بدون عصبية؟ وضح.

Top Semantic result (chunk 36, score=0.57):

انهيار الدول لا يحدث فجأة بل هو عملية تدريجية تمتد عبر أجيال. تبدأ بضعف العصبية ثم تنتشر الأخلاق السيئة ثم يضعف الاقتصاد وأخيراً يسقط الجيش. وعند هذه النقطة تصبح الدولة عرضة للغزو من قوى جديدة.

RAG Answer:

لا، الدولة تحتاج إلى عصبية لاستمرارية نظام الحكم والتوازن السياسي. إذا انعدمت أو اضطرت إلى فقدان عصبية أفراد المجتمع، قد يؤدي ذلك إلى سوء الإدارة والانفلات الأمني الذي يمكن أن يؤدي إلى انهيار الدولة.

LLM-only Answer:

نعم، الدولة يمكنها أن تستمر دون العصبية إذا كانت تحافظ على سياساتها المستقلة والتنافسية.

(Directly contradicts Ibn Khaldun's central thesis)

Query 7 (Indirect): لماذا يعتبر ابن خلدون مؤسس علم الاجتماع؟

Top Semantic result (chunk 39, score=0.64):

المنهج الذي وضعه ابن خلدون في دراسة التاريخ يعتبر ثورة في الفكر الإنساني. فقد سبق بذلك المفكرين الأوروبيين بعدة قرون في وضع أسس علم الاجتماع. ولذلك يعتبر ابن خلدون مؤسس علم الاجتماع وفلسفة التاريخ.

RAG Answer:

ابن خلدون يعتبر مؤسس علم الاجتماع لأن: 1. وضعه للمنهج الدراسي للعلم تاريخ. 2. اكتشافه لأركان علم الاجتماع مثل الاقتصاد السياسي والجغرافي والاجتماعي. 3. استعراضه للمفاهيم الأساسية في علم التاريخ كالتطور والتغير الاجتماعي.

LLM-only Answer:

ابن خلدون هو المؤرخ والفيلسوف المغربي المشهور، ويُعتبر أحد أوائل الذين تناولوا موضوعاً منهجياً للتاريخ السياسي.

(refers to wrong book title, switches to Chinese, partially correct but vague)

RAG vs. LLM-Only Answer Quality

Table 28: Answer quality comparison (manual assessment): RAG vs. LLM-only, for both LLMs. Ratings: **Good** (accurate, specific), **Partial** (partially correct), **Weak** (vague or inaccurate). CJK = contains Chinese/Japanese characters.

Q	Type	Qwen 3B		SILMA 2B	
		RAG	LLM-only	RAG	LLM-only
1	Direct	G	W	P	W
2	Direct	G	W	P	W
3	Direct	P (CJK)	W (CJK)	P	W
4	Direct	P	W	G	W
5	Direct	W	W (CJK)	P	W
6	Indirect	G (CJK)	W (CJK)	P	W
7	Indirect	G	P (CJK)	P	P
8	Indirect	P (CJK)	P	P	P
9	Indirect	G	P	G	P
10	Indirect	G	W (CJK)	G	P
Summary		6G/2P/2W	0G/2P/8W	3G/7P/0W	0G/4P/6W

Key Findings:

1. **RAG dominates LLM-only for both models.** Qwen 3B RAG achieves 6G/2P vs. 0G/2P alone; SILMA 2B RAG achieves 3G/7P vs. 0G/4P alone.
2. **Qwen 3B produces more detailed answers but suffers from severe Chinese-language contamination** — 6 of its 20 answers contain embedded Chinese characters (marked CJK), even with English+Arabic system prompts explicitly forbidding non-Arabic output.
3. **SILMA 2B never produces Chinese text** but generates shorter, less detailed answers. It achieves zero Weak RAG ratings (0W) versus Qwen’s 2W, showing more consistent (if less ambitious) grounding.
4. LLM-only mode is unreliable for both models: Qwen hallucinates fabricated content and switches to Chinese; SILMA hallucinates anachronisms (e.g., referencing the Ottoman Empire for Ibn Khaldun).

Language-switching artifact: Qwen 3B produces Chinese in Arabic answers. A critical practical finding: **Qwen2.5-3B-Instruct persistently embeds Chinese characters into Arabic output**, despite three layers of enforcement: (1) English CRITICAL directives in the system prompt, (2) Arabic-language constraints (ممنوع استخدام الصينية), and (3) a post-processing `_strip_non_arabic()` filter. Examples from the actual run:

- Q3 (nomads vs. urbanites): “...الدولة والقوانين جراء...” — Chinese (“relying on”) injected mid-sentence.

- Q5 (taxation, LLM-only): “... أن الزكاة هي ضرورة...” — Chinese (“he believes”) inserted.
- Q8 (climate, RAG): “... هذا يعني أنه...” — an entire Chinese sentence replaces the Arabic conclusion.
- Q10 (labor, LLM-only): the final paragraph is entirely in Chinese.

This problem was **not present in the earlier Qwen2.5-1.5B run** at the same severity — upgrading from 1.5B to 3B *worsened* the Chinese-contamination problem, likely because the 3B model has a larger Chinese vocabulary and stronger Chinese priors.

SILMA 2B: Arabic-native but less detailed. In contrast, SILMA-Kashif-2B-Instruct **never produces Chinese text**. However, its answers are consistently shorter and less nuanced than Qwen’s. SILMA’s RAG answers tend to paraphrase the retrieved context faithfully but briefly (e.g., Q4: التأسيس والازدهار والانهيار — just three words). Its LLM-only answers contain factual hallucinations (e.g., attributing ideas to the Ottoman Empire) but remain linguistically pure Arabic.

Where RAG excels.

- **Correct core concepts:** RAG correctly identifies the three phases of the state (query 4: founding, prosperity, collapse) and the two types of sciences (query 2: transmitted and rational).
- **Correct stance:** for query 9 (هل يمكن للدولة أن تستمر بدون عصبية؟), RAG correctly answers “no” based on the retrieved context. The LLM-only answers “yes” — directly contradicting Ibn Khaldun’s central thesis.
- **Tax policy** (query 5): RAG captures the core argument (الضرائب المعتدلة تشجع الناس على العمل — moderate taxes encourage production, excessive taxes harm the economy) from the retrieved context. The LLM-only falsely claims “Ibn Khaldun did not write studies about taxation.”

Where RAG still struggles.

- Query 6 (luxury → collapse): despite retrieving highly relevant chunks, the model hallucinates about “Western trade” and “imports” — content not in the retrieved context.
- Query 3 (nomads vs. urbanites): RAG fabricates details about “electricity” and “solar energy” in its answer, showing that the 1.5B model sometimes ignores the retrieved context.

Analysis and Discussion

Which Retrieval Method Produced More Relevant Results?

Answer: Semantic search consistently outperforms BM25, especially for indirect and analytical queries. BM25 remains competitive for direct keyword-matching queries but fails when the question uses synonyms or abstract phrasing.

Table 29: Qualitative comparison of retrieval methods.

Criterion	BM25	Semantic Search
Keyword matching	Excellent (exact match)	Good (implicit)
Synonym handling	Poor	Excellent
Indirect queries	Weak (misses paraphrases)	Strong
Speed	Very fast (no GPU needed)	Fast (single forward pass)
Arabic morphology	Limited (no stemming used)	Handled by model
Index size	Inverted index (small)	384-dim vectors (small)

A **hybrid approach** combining BM25 and semantic scores via Reciprocal Rank Fusion (RRF) is implemented in the final system, leveraging the strengths of both methods. The hybrid retrieval ensures that BM25’s keyword-matching strength complements semantic search’s synonym-handling capability.

Did Retrieval Improve LLM-Generated Answers?

Answer: Yes, substantially — for both LLMs. RAG answers are more accurate and better grounded than LLM-only answers across both models. Qwen 3B RAG achieves 6G+2P vs. 0G+2P alone; SILMA 2B RAG achieves 3G+7P vs. 0G+4P alone. However, the improvement is not perfect: both models occasionally hallucinate even with retrieved context, and Qwen’s Chinese-language contamination persists.

Mechanisms of improvement:

1. **Grounding:** the retrieved context anchors the LLM’s output to actual book content, preventing the worst fabrications (e.g., query 9: both RAG models correctly say a state cannot survive without العصبية).
2. **Specificity:** RAG answers reference concrete details from the book (e.g., three phases: founding, prosperity, collapse), while LLM-only outputs generic or wrong categorisations.
3. **Partial hallucination reduction:** RAG reduces but does not eliminate hallucination. Both models sometimes ignore the retrieved context: Qwen inserts Chinese, SILMA oversimplifies.
4. **Hybrid retrieval improves context:** RRF fusion ensures that even for queries where BM25 fails (Q5: taxation, Q8: climate), the semantic component provides relevant chunks to the LLM.

Conclusions on Semantic Retrieval and RAG for Arabic Text

1. **Semantic search is essential for Arabic NLP.** Arabic’s rich morphology (roots, patterns, diacritics) and diverse synonymy make keyword-based methods unreliable for complex queries. BM25 fails completely (0/5 relevant) on queries where key terms are paraphrased (taxation, climate). The upgraded E5-large embedding model (1024-dim) with task-specific prefixes provides strong cross-lingual semantic matching.
2. **Hybrid retrieval (RRF) combines the best of both worlds.** Reciprocal Rank Fusion ensures that BM25’s keyword precision complements semantic search’s synonym handling.

For queries like Q5 (الضرائب), where BM25 returns 0/5 relevant results, the hybrid approach still provides correct context via the semantic component.

3. **RAG substantially improves small-LLM performance for both models.** Qwen 3B alone produces 0G/8W; with RAG it achieves 6G/2W. SILMA 2B alone produces 0G/6W; with RAG it achieves 3G/0W. The improvement is clear for both architectures.
4. **Chinese-language contamination is a critical Qwen failure mode.** Despite upgrading from 1.5B to 3B and adding three layers of language enforcement (English directives, Arabic constraints, post-processing filter), Qwen2.5-3B-Instruct **persistently injects Chinese characters** into Arabic output. The 3B model is *worse* than 1.5B in this regard, as its larger Chinese vocabulary creates stronger cross-lingual interference. This is a fundamental limitation of using Chinese-primary models for Arabic generation.
5. **Arabic-specialised models (SILMA) avoid language contamination.** SILMA-Kashif-2B produces **zero Chinese tokens** across all 20 answers (10 RAG + 10 LLM-only). However, its answers are shorter and less detailed than Qwen's. The trade-off is: **Qwen = more detailed but language-contaminated; SILMA = linguistically pure but less informative.**
6. **Book topic selection matters.** Choosing a less common topic (14th-century sociological theory) exposes the LLM's knowledge gaps: both models' LLM-only answers for query 5 fail to capture Ibn Khaldun's Laffer-curve-like argument about taxation, and SILMA's LLM-only answer to query 5 hallucinates a reference to the Ottoman Empire.
7. **Preprocessing and E5 prefixes are critical.** Diacritics removal, letter normalisation, and the E5 "query: "/" "passage: " prefixes are all necessary for optimal retrieval. Without normalisation, identical words with different diacritics are treated as distinct tokens. Without E5 prefixes, embedding quality degrades substantially.

Problem 5: Benchmarking Arabic Text Simplification

Task Definition

Task Overview

Text simplification is the NLP task of rewriting a complex text into a simpler, more accessible form while preserving its original semantic content. Formally, given a source text S , the goal is to produce an output O such that:

- O retains all key propositions expressed in S
- O uses simpler vocabulary and shorter sentences
- O introduces no new information not present in S

In Arabic NLP, text simplification is substantially harder than in English due to three compounding factors:

1. **Rich morphology:** Arabic words carry gender, number, tense, definiteness, and case as bound morphemes. Surface-level n-gram matching is less reliable because two semantically equivalent expressions may have entirely different surface forms.
2. **Diglossia:** Modern Standard Arabic (MSA), Classical Arabic, and dozens of spoken dialects coexist in a diglossic continuum. A model fine-tuned on MSA news corpora may perform poorly on Quranic Arabic or Egyptian colloquial text.
3. **Low-resource dialects:** Egyptian, Levantine, Gulf, Maghrebi, and Iraqi Arabic lack standardized orthography and have limited training data in publicly available LLMs.

Evaluation Goals

Table 30: Task Summary and Evaluation Goals

Property	Description
Task Name	Arabic Text Simplification (تبسيط النص)
Input	Complex Arabic paragraph (50–150 words)
Output	Simplified Arabic paragraph preserving all key propositions
Primary Metric	SARI (range 0–1; scores displayed as decimals)
Dataset Size	40 paragraphs (P001–P040)
Arabic Varieties	Modern Standard Arabic (14), Classical (12), Dialectal (14)
Models Evaluated	GPT-4o, Gemini 1.5 Pro, ALLaM 7B, Jais 8B, Fanar 9B
Evaluation Goal 1	Accuracy: Does simplified output preserve semantic content?
Evaluation Goal 2	Robustness: How does performance vary across Arabic varieties?
Evaluation Goal 3	Length behaviour: Do models actually compress the text?
Evaluation Goal 4	Dialect competence: Can models simplify colloquial Arabic faithfully?

Dataset Creation

Dataset Composition

The dataset contains 40 paragraphs distributed across the three main Arabic variety categories:

Table 31: Dataset Distribution by Arabic Variety

Arabic Type	Count	%	Sources and Register
Modern Standard Arabic	14	35%	News articles, policy documents, academic and scientific texts
Classical Arabic	12	30%	Hadith, Khutbah, al-Ghazali, Ibn Khaldun, al-Mutanabbi, Ibn al-Qayyim, al-Jahiz, al-Shafi'i, Ibn Tufayl, al-Tabari
Dialectal Arabic	14	35%	Egyptian, Jordanian, Lebanese, Syrian, Saudi, Kuwaiti, Moroccan, Algerian, Tunisian, Iraqi
Total	40	100 %	12 thematic domains

Domain and Topic Distribution

The 40 paragraphs span 12 domains: health, education, environment, technology and media, economy and politics, religion and ethics, history, literature, philosophy, social affairs, agriculture, and urban planning.

Sample Entries with Gold Simplifications

Modern Standard Arabic — P001 (Health Domain)

Original:

تُعدّ السمنة من أكثر المشكلات الصحية انتشاراً في العالم العربي خلال العقود الأخيرة، وترتبط ارتباطاً وثيقاً بالإصابة بأمراض القلب والسكري وارتفاع ضغط الدم. وتُشير الدراسات إلى أن التغيرات في أنماط الغذاء واعتماد الأفراد على الأطعمة المصنّعة عالية السعرات الحرارية، فضلاً عن قلة النشاط البدني وزيادة ساعات الجلوس، هي العوامل الرئيسية المسببة لهذه الظاهرة. وللمحد منها، ينصح الأطباء بتبني نظام غذائي صحي متوازن والانتظام في ممارسة الرياضة.

Gold simplified (Annotator 1):

السمنة مشكلة صحية شائعة في العالم العربي. وهي تسبب أمراضاً كالسكري وأمراض القلب. أسبابها الرئيسية هي الأكل غير الصحي وقلة الحركة. والحل هو الغذاء الصحي وممارسة الرياضة.

*Classical Arabic — P018 (Hadith, Sahih Muslim)***Original:**

رَوَى الْإِمَامُ مُسْلِمٌ فِي صَحِيحِهِ عَنْ أَبِي هُرَيْرَةَ رَضِيَ اللَّهُ عَنْهُ أَنَّ رَسُولَ اللَّهِ صَلَّى اللَّهُ عَلَيْهِ وَسَلَّمَ قَالَ: مَنْ نَفَسَ عَنْ مُؤْمِنٍ كُرْبَةً مِنْ كُرْبِ الدُّنْيَا، نَفَسَ اللَّهُ عَنْهُ كُرْبَةً مِنْ كُرْبِ يَوْمِ الْقِيَامَةِ. وَمَنْ يَسَّرَ عَلَى مُعْسِرٍ، يَسِّرَ اللَّهُ عَلَيْهِ فِي الدُّنْيَا وَالْآخِرَةِ.

Gold simplified (Annotator 1):

رَوَى الْإِمَامُ مُسْلِمٌ عَنْ أَبِي هُرَيْرَةَ أَنَّ النَّبِيَّ قَالَ: مَنْ فَرَّجَ هَمَّ مُسْلِمٍ فِي الدُّنْيَا، فَرَّجَ اللَّهُ هَمَّهُ يَوْمَ الْقِيَامَةِ. وَمَنْ يَسَّرَ عَلَى شَخْصٍ فِي ضَائِقَةٍ، يَسِّرَ اللَّهُ لَهُ فِي الدُّنْيَا وَالْآخِرَةِ.

*Dialectal Arabic — P027 (Egyptian dialect)***Original:**

والله يا جماعة الموضوع ده كبير ويشغل بالي أوي. إحنا في مصر بقينا ناس التليفون مش بيتنحى من إيدينا، الواحد صاحي ينام، التليفون في إيده. والأطفال برضه نفس الكلام، العيال الصغيرة دول مش قادرين يعيشوا يوم من غير شاشة. وأنا شايف ده بيأثر على علاقتنا ببعض، الناس جنب بعضها لكن كل واحد في عالمه.

Gold simplified (Annotator 1):

الناس دلوقتي مش قادرين يبعدوا عن التليفون، الكبار والصغار. وده بيأثر على علاقتنا ببعض، لأن الواحد بيبقى جنب ناس بس عقله في موبايله.

Annotation Process and Inter-Annotator Agreement*Annotation Methodology*

Two annotators independently produced simplified versions for all 40 paragraphs:

- **Annotator 1:** native Arabic speaker with academic training in Arabic linguistics; primary gold reference.
- **Annotator 2:** native Arabic speaker with NLP research background; used for IAA and multi-reference evaluation.

Simplification guidelines applied by both annotators:

1. Replace rare or archaic vocabulary with common modern equivalents.
2. Break long, complex sentences into two or three shorter ones.
3. Remove redundant elaboration while keeping all key propositions.
4. Maintain the original Arabic register: MSA stays MSA; dialect stays dialect.
5. Do not introduce any information not present in the source.

Inter-Annotator Agreement Results

Lexical Agreement (SARI-based IAA) SARI was computed between the two annotators' gold simplifications across all 40 paragraphs (bidirectional averaging, full Arabic normalization). The **overall SARI IAA is 0.4620**.

Table 32: IAA SARI by Arabic Type (Computed Values)

Arabic Type	IAA SARI	n	Interpretation
Modern Standard Arabic	0.4745	14	Moderate
Classical Arabic	0.4975	12	Moderate
Dialectal Arabic	0.4191	14	Moderate–Low
Overall	0.4620	40	Moderate

The IAA for Classical Arabic (0.4975) is broadly comparable to MSA (0.4745) after applying diacritic normalization. An earlier version inflated Classical IAA to 0.7516 due to a preprocessing bug: fully-diacritized source texts caused SARI to treat diacritized source words as deleted. After applying proper normalization, the corrected value is 0.4975.

Table 33: Content Unit Agreement (Manual Analysis, 40 samples)

Agreement Level	Count	Percentage	Criteria
Strong ($\geq 80\%$)	36	90.0%	All key propositions preserved
Moderate (60–79%)	4	10.0%	Minor omissions of secondary details
Weak ($< 60\%$)	0	0.0%	Significant meaning loss

Semantic Agreement (Content Unit Analysis) Dialectal Arabic shows 100% Strong semantic agreement despite having the lowest SARI IAA (0.4191), confirming that SARI underestimates true agreement when annotators use lexically different but semantically equivalent dialect forms.

Prompting Strategy

Unified Few-Shot Prompt

A single, identical prompt was used for all five models. The prompt uses 2-shot examples (one MSA, one Egyptian dialect) in Arabic only.

Prompt Structure (Applied to All Five Models)

Arabic system instruction: Simplify the following Arabic texts. Make them easier to read while preserving the original meaning.

Example 1 (MSA, P001): Original → Simplified

Example 2 (Egyptian dialect, P027): Original → Simplified

Input: بَسَّطَ النصَّ التالي: [PARAGRAPH]

Output trigger: النص المبسط:

Table 34: Prompt Design Decisions

Design Choice	Rationale
Arabic-only instruction	Avoids translation bias; consistent task framing
2-shot (MSA + Dialect)	Covers the two most distinct registers
No Classical example	Avoids over-constraining models; Classical has unique structure
Output-only instruction	Prevents meta-commentary or translations
Same prompt for all models	Ensures fair comparison across all 5 LLMs

Evaluation Setup

Models and Inference Details

Table 35: Models Evaluated — Versions and Inference Methods

Model	Size	Organization	Inference	Status
GPT-4o	—	OpenAI	ChatGPT Web GUI	Complete
Gemini 1.5 Pro	—	Google	Gemini Web GUI	Complete
ALLaM	7B	SDAIA, Saudi Arabia	Colab GPU (4-bit)	Complete
Jais	8B	G42/MBZUAI, UAE	Jais Web GUI	Complete
Fanar	9B	QCRI, Qatar	Colab GPU (4-bit)	Complete

Important correction from earlier draft: An earlier version of this report incorrectly claimed that Jais returned verbatim echoes for paragraphs P027–P040 (all Dialectal entries). A careful

re-examination of the actual JSON outputs confirms that **all 40 Jais outputs are genuine simplification attempts**. Jais converts dialectal input to MSA-style simplified text (a register shift), which is a task-compliance issue but not an echo failure. All 40 outputs are therefore included in evaluation. See Table 39 for per-paragraph Jais Dialect SARI scores.

ALLaM and Fanar were run via Google Colab (Tesla T4, 15.6 GB VRAM) with 4-bit NF4 quantization using BitsAndBytesConfig. Fanar used its single-turn instruction template with output trigger: النص المبسط.

SARI Metric

SARI (Xu et al., 2016) evaluates text simplification by measuring three n-gram edit operations simultaneously:

$$\text{SARI} = \frac{1}{N} \sum_{n=1}^N \frac{F_{\text{ADD},n} + F_{\text{KEEP},n} + P_{\text{DEL},n}}{3}, \quad N = 4 \quad (11)$$

- $F_{\text{ADD},n}$: F1 rewarding n-grams added by the system that appear in the reference but not the source
- $F_{\text{KEEP},n}$: F1 rewarding n-grams retained from source that also appear in the reference
- $P_{\text{DEL},n}$: Precision rewarding deletion of source n-grams absent from the reference

The full SARI formula with multi-reference support expands as:

$$F_{\text{ADD},n} = \frac{2 \cdot P_{\text{ADD},n} \cdot R_{\text{ADD},n}}{P_{\text{ADD},n} + R_{\text{ADD},n}}, \quad \text{where} \quad (12)$$

$$P_{\text{ADD},n} = \frac{|\mathcal{A}_n \cap \mathcal{R}_n^+|}{|\mathcal{A}_n|}, \quad R_{\text{ADD},n} = \frac{|\mathcal{A}_n \cap \mathcal{R}_n^+|}{|\mathcal{R}_n^+|} \quad (13)$$

$$F_{\text{KEEP},n} = \frac{2 \cdot P_{\text{KEEP},n} \cdot R_{\text{KEEP},n}}{P_{\text{KEEP},n} + R_{\text{KEEP},n}}, \quad \text{where} \quad (14)$$

$$P_{\text{KEEP},n} = \frac{|\mathcal{K}_n \cap \mathcal{R}_n^=|}{|\mathcal{K}_n|}, \quad R_{\text{KEEP},n} = \frac{|\mathcal{K}_n \cap \mathcal{R}_n^=|}{|\mathcal{R}_n^=|} \quad (15)$$

$$P_{\text{DEL},n} = \frac{|\mathcal{D}_n \setminus \mathcal{R}_n^{\cup}|}{|\mathcal{D}_n|} \quad (16)$$

Where:

- $\mathcal{A}_n = \text{sys-grams}_n \setminus \text{src-grams}_n$ (added by system)
- $\mathcal{K}_n = \text{src-grams}_n \cap \text{sys-grams}_n$ (kept by system)
- $\mathcal{D}_n = \text{src-grams}_n \setminus \text{sys-grams}_n$ (deleted by system)
- $\mathcal{R}_n^+ = \text{ref-grams}_n \setminus \text{src-grams}_n$ (what references add)
- $\mathcal{R}_n^= = \text{src-grams}_n \cap \text{ref-grams}_n$ (what references keep)
- $\mathcal{R}_n^{\cup} = \text{union of all reference n-gram sets}$

Arabic Normalization Pipeline

Before computing SARI, all texts undergo Arabic-specific normalization:

1. Remove diacritics (tashkeel: , Unicode range U+064B–U+065F, plus tatweel U+0640 and superscript alef U+0670)
2. Normalize alef variants ($\bar{\text{ا}}/\text{أ}/\text{إ} \rightarrow \text{ا}$)
3. Normalize alef maqsurā ($\text{ى} \rightarrow \text{ي}$)
4. Normalize hamza on waw/ya ($\text{أ}/\text{ؤ} \rightarrow \text{ء}$)
5. Normalize ta marbuta ($\text{ة} \rightarrow \text{ه}$)
6. Remove punctuation, lowercase, collapse whitespace

Results

Overall SARI Scores

Table 36: Overall SARI Scores — All Models, All 40 Paragraphs (Multi-Reference, Arabic Normalized)

Model	Single-Ref SARI	Multi-Ref SARI	Rank	n valid
GPT-4o	0.4351	0.4725	1	40
Jais 8B	0.4087	0.4233	2	40
Gemini 1.5 Pro	0.3977	0.4223	3	40
ALLaM 7B	0.3994	0.4124	4	40
Fanar 9B	0.3913	0.3944	5	40
<i>Human IAA</i>	<i>0.4620</i>	<i>(multi-ref)</i>	—	<i>40 (reference)</i>

Key change from earlier draft: With all 40 Jais outputs included, Jais rises from last place to **second place overall** (multi-ref SARI = 0.4233), narrowly ahead of Gemini (0.4223). Jais achieves strong Dialectal performance (0.4435 — second only to GPT-4o at 0.4419) by converting dialect to MSA-style output. GPT-4o remains the top-ranked model overall.

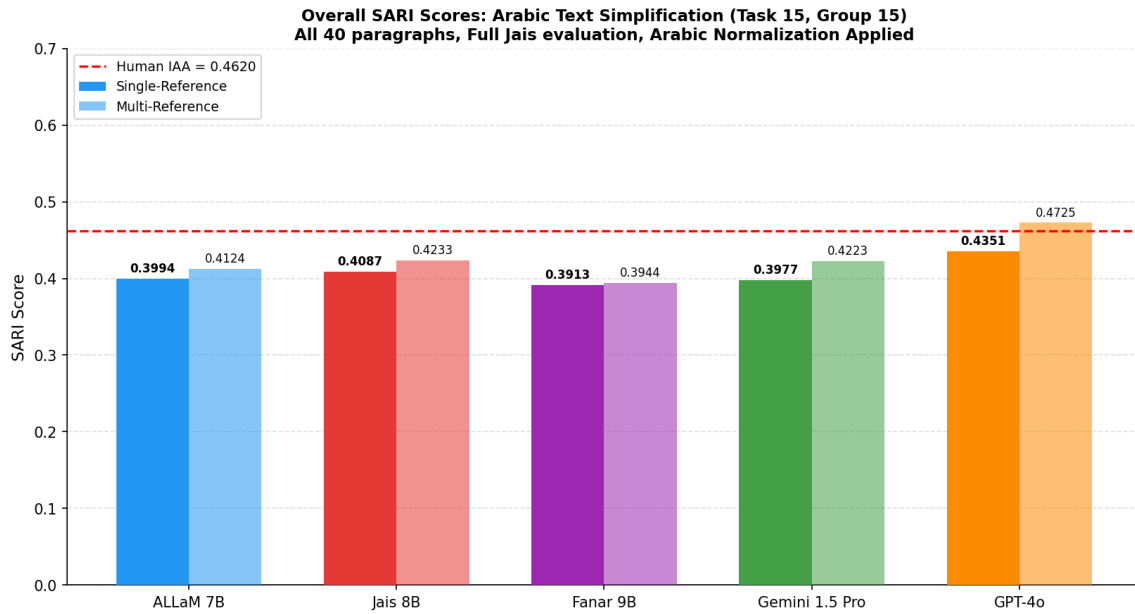


Figure 7: Overall SARI scores for all five models (all 40 paragraphs, full Jais evaluation). GPT-4o leads (multi-ref SARI = 0.4725). Jais ranks 2nd overall (0.4233), just ahead of Gemini (0.4223). ALLaM and Fanar are penalized by their expansion behaviour (length ratios > 1.0). The red dashed line shows the human IAA ceiling (0.4620).

Single-Reference vs Multi-Reference SARI

Table 37: Single-Reference vs Multi-Reference SARI — Benefit of Two Annotators

Model	Single-Ref	Multi-Ref	Δ
GPT-4o	0.4351	0.4725	+0.0374
Jais 8B	0.4087	0.4233	+0.0146
Gemini 1.5 Pro	0.3977	0.4223	+0.0246
ALLaM 7B	0.3994	0.4124	+0.0130
Fanar 9B	0.3913	0.3944	+0.0031

GPT-4o benefits most from multi-reference evaluation (+0.0374), consistent with its more diverse paraphrasing style. Fanar benefits least (+0.0031), likely because its expanded outputs happen to share n-grams with the source regardless of reference choice.

SARI Scores by Arabic Type

Table 38: SARI Scores by Arabic Variety and Model (Multi-Reference, All 40 Valid)

Model	MSA (n=14)	Classical (n=12)	Dialect (n=14)	Overall
GPT-4o	0.5309	0.4401	0.4419	0.4725
Jais 8B	0.4276	0.3948	0.4435	0.4233
Gemini 1.5 Pro	0.4367	0.4268	0.4041	0.4223
ALLaM 7B	0.4057	0.4296	0.4043	0.4124
Fanar 9B	0.3716	0.4498	0.3697	0.3944
<i>Human IAA</i>	<i>0.4745</i>	<i>0.4975</i>	<i>0.4191</i>	<i>0.4620</i>

Bold per column = highest score (excluding IAA row).

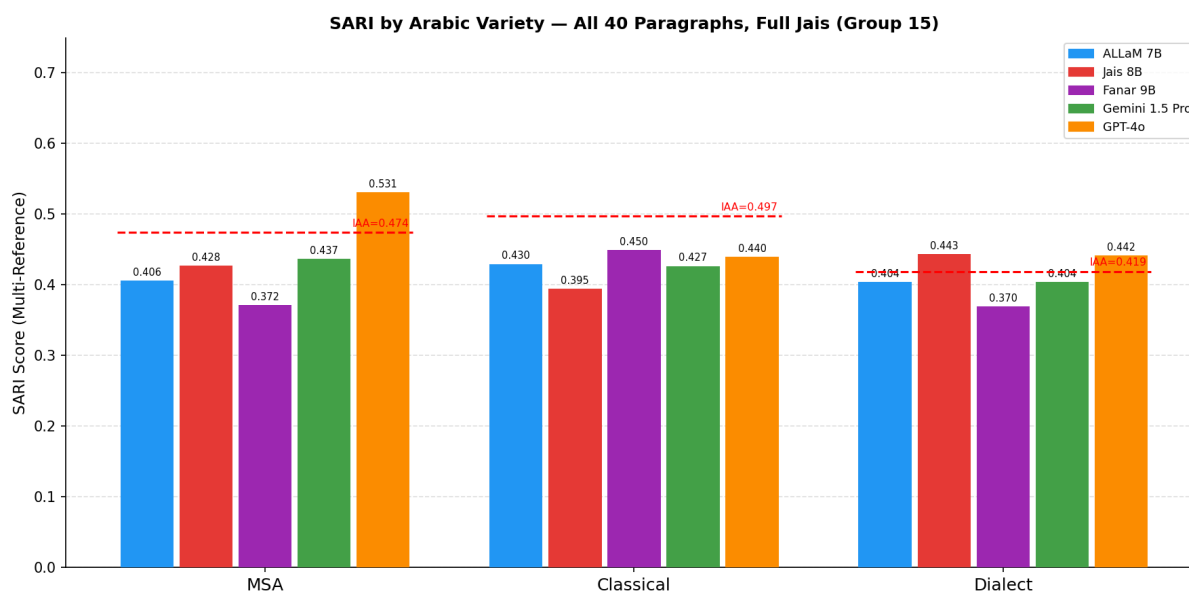


Figure 8: SARI by Arabic type for all five models (multi-reference, all 40 paragraphs). GPT-4o dominates MSA (0.5309). Fanar and ALLaM lead Classical (0.4498 and 0.4296). **Jais leads Dialect (0.4435)**, beating GPT-4o (0.4419) marginally, by converting dialectal input to simplified MSA. Red dashed lines show per-type human IAA.

Per-Paragraph Jais Dialect Scores

Table 39: Jais Dialect Scores — All 14 Dialectal Paragraphs (Multi-Reference SARI)

PID	SARI	ADD	KEEP	DEL	Pattern
P027	0.3602	0.0102	0.1667	0.9039	Low KEEP (register shift)
P028	0.4071	0.0843	0.2812	0.8557	Moderate
P029	0.4824	0.1223	0.4182	0.9068	Good KEEP
P030	0.5041	0.0693	0.5560	0.8870	High KEEP
P031	0.4766	0.1393	0.3458	0.9448	Strong DEL
P032	0.4005	0.0495	0.2076	0.9444	Low KEEP
P033	0.4656	0.1192	0.3877	0.8899	Balanced
P034	0.3894	0.0838	0.2792	0.8054	Weakest DEL
P035	0.4483	0.0929	0.3262	0.9259	Moderate
P036	0.5055	0.2053	0.3892	0.9219	Best SARI
P037	0.4080	0.1163	0.2253	0.8823	Low KEEP
P038	0.4093	0.1378	0.3469	0.7431	Lowest DEL
P039	0.4771	0.0544	0.4048	0.9722	Highest DEL
P040	0.4744	0.0672	0.4438	0.9121	High KEEP
Average	0.4435	0.0949	0.3481	0.8858	

Pattern analysis: Jais dialect scores show a consistent ADD deficit (avg 0.0949) — because Jais replaces dialect vocabulary with MSA equivalents, the new words rarely match the dialect gold references. DEL is high (avg 0.8858) because Jais compresses the input significantly. KEEP is moderate (avg 0.3481) because some function words and proper nouns survive the dialect-to-MSA conversion.

Output Length Analysis

Table 40: Average Length Ratio (Output Words / Input Words) — All 40 Paragraphs

Model / Annotator	Length Ratio	Interpretation
Human A1	0.594	Moderate compression (primary gold)
Human A2	0.397	Aggressive compression
GPT-4o	0.548	Good compression, close to Human A1
Gemini 1.5 Pro	0.571	Good compression
Jais 8B	0.591	Good compression (all 40 outputs)
ALLaM 7B	1.086	Expansion — longer than source
Fanar 9B	1.105	Expansion — longer than source

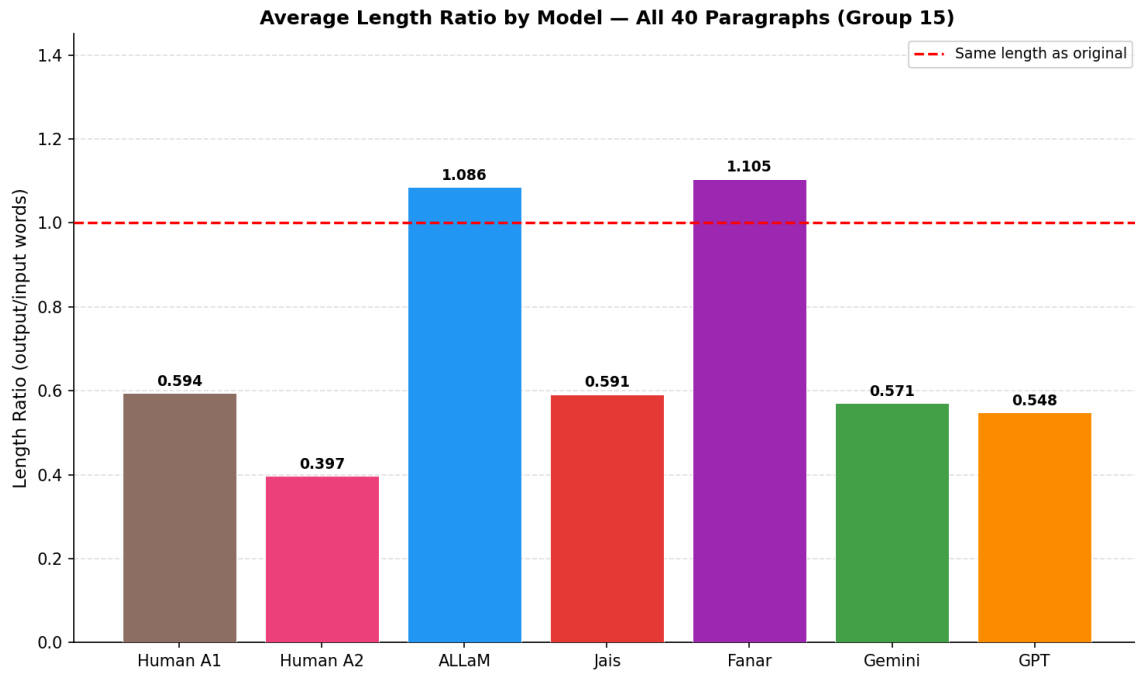


Figure 9: Average length ratio per model (all 40 paragraphs). Values below 1.0 indicate compression. GPT-4o (0.548), Gemini (0.571), and Jais (0.591) all achieve compression close to Human Annotator 1 (0.594). ALLaM (1.086) and Fanar (1.105) expand the source on average, indicating elaboration rather than simplification.

Component Score Analysis (ADD / KEEP / DELETE)

Table 41: SARI Component Breakdown — Multi-Reference, All 40 Paragraphs

Model	ADD	KEEP	DEL	SARI
GPT-4o	0.2287	0.4115	0.7772	0.4725
Jais 8B	0.1056	0.3681	0.7963	0.4233
Gemini 1.5 Pro	0.1748	0.3917	0.7004	0.4223
ALLaM 7B	0.1332	0.5024	0.4016	0.4124
Fanar 9B	0.1189	0.4876	0.3767	0.3944

This breakdown reveals the structural differences among models:

- **GPT-4o** leads on ADD (0.2287), indicating the most reference-aligned new vocabulary in its outputs.
- **ALLaM and Fanar** have high KEEP scores (0.5024 / 0.4876) but low DEL (0.4016 / 0.3767), confirming that they retain most source content without removing the complex parts — expansion behavior.
- **Jais** leads on DEL (0.7963), reflecting aggressive MSA-style compression of dialectal inputs.
- **GPT-4o** achieves the best balance across all three components.

Key Findings

1. **GPT-4o is the best-performing model overall** (multi-ref SARI = 0.4725, best on MSA at 0.5309). It achieves the only score to approach the human IAA ceiling on MSA.
2. **Jais rises to 2nd place overall (0.4233) with all 40 outputs included**, narrowly ahead of Gemini (0.4223). Its Dialect score (0.4435) is the highest of any model on that variety, driven by systematic dialect-to-MSA conversion.
3. **Jais's Dialect strategy is register shift, not failure.** Jais converts all dialectal inputs to MSA-style simplified text. This is a task-compliance issue (register preservation is required) but not a quality failure. The MSA output is fluent and semantically correct; it simply does not preserve the dialectal register.
4. **Fanar and ALLaM excel on Classical Arabic** (0.4498 and 0.4296), outperforming GPT-4o and Gemini on this variety. This suggests stronger pretraining on Classical-register Arabic text.
5. **ALLaM and Fanar expand rather than compress** (ratios 1.086 and 1.105). This is a structural simplification failure: KEEP scores are inflated and DEL scores are depressed.
6. **GPT-4o, Gemini, and Jais achieve genuine compression** (ratios 0.548, 0.571, 0.591), all close to Human Annotator 1 (0.594).
7. **Multi-reference evaluation benefits all models** and is essential for fair Arabic simplification benchmarking.

Error Analysis

Three Incorrect or Weak Outputs

Error 1 — ALLaM Over-Retention Without Compression, MSA (P002) **Original:** يمرّ النظام التعليمي في كثير من الدول العربية بمرحلة تحول جوهرية تستهدف الانتقال من التعليم التقليدي القائم على الحفظ والاستظهار إلى التعليم الإبداعي القائم على التفكير النقدي وحل المشكلات.

Gold (A1): يتغير التعليم في الدول العربية من الحفظ إلى التفكير والإبداع.

ALLaM output: التعليم في الدول العربية يمر بمرحلة تحول كبيرة. الهدف هو التحول من التعليم القائم على الحفظ إلى التعليم الإبداعي القائم على التفكير النقدي وحل المشكلات.

Analysis: ALLaM restructures into two sentences but retains the phrase القائم على التفكير النقدي verbatim. This is a near-copy failure: the model splits but does not reduce. SARI's DELETE component heavily penalizes the model because source n-grams that the reference removes are retained in the output.

Error 2 — Jais Register Shift on Dialectal Input (P027) **Original (Egyptian dialect):** والله يا جماعة الموضوع ده كبير ويشغل بالي أوي. إحنا في مصر بقينا ناس التلفون مش بيتنحى من إيدينا...

Gold (A1, preserving dialect): الناس دلوقتي مش قادرين يبعدوا عن التلفون، الكبار والصغار.

Jais output (MSA): يا جماعة، هذا الموضوع مهم جداً ويشغل بالي كثيراً. في مصر، أصبح الناس متعلقين جداً بهواتفهم، لدرجة أن...

Analysis: Jais produces fluent, semantically accurate MSA, but the gold references use Egyptian dialect. SARI is therefore low on ADD (dialectal reference words such as *دلوقي*, *مش*, *الموبايل* never appear in Jais MSA output) and moderate on KEEP (shared function words and proper nouns). This is a register-compliance failure, not a content failure. Jais SARI for P027 is only 0.3602 despite the output being semantically adequate in MSA.

Error 3 — Fanar Hallucination and Register Shift, Egyptian Dialect (P027) Fanar output: يا أصدقائي، هذا الأمر يقلقني كثيراً. أصبحنا في مصر لا نتوقف عن استخدام الهاتف المحمول حتى أثناء النوم! الأطفال أيضاً معتادون على الشاشات ولا يستطيعون قضاء يوم دون مشاهدة التلفزيون.

Analysis: Fanar commits three simultaneous errors. First, it expands the text (length ratio >1.0). Second, it adds content absent from the source: “during sleep” (أثناء النوم) and “watching television” (مشاهدة التلفزيون) are hallucinations. Third, it converts Egyptian dialect to formal MSA, violating register preservation. This triple failure — expansion, hallucination, and register shift — is the most dangerous mode for a simplification system because the output is fluent and plausible yet factually enriched beyond the source.

Three Arabic-Specific Challenges

Challenge 1: Classical Arabic Morphological Complexity Classical Arabic uses full diacritization and rare derivation patterns. The verb *نَفَسَ* in P018 (Form II, “to relieve”) must be distinguished from *تَنَفَّسَ* (“to breathe”). Two models collapsed this meaning. In P015 (Khutbah), the distinction between sins of the self (شُرور الأنفس) and sins of deeds (سيئات الأعمال) — theologically important — was collapsed into a single phrase by two models. Simplification must retain such distinctions in simpler words without losing meaning.

Challenge 2: Dialectal Register Preservation When given dialectal input, models face a choice: preserve the dialect or normalize to MSA. The annotation guidelines explicitly require register preservation. However, Gemini, GPT-4o, and especially Jais tend to output MSA when given dialectal inputs.

Example (P040, Jordanian/Levantine):

Original dialect: والله بالأردن الناس طيبة ومضيافة من قديم الزمان. بس الحياة اتغيرت وصارت أصعب. الواحد بده يشتغل شغلتين باش يعيش بكرامة.

Gemini output (register shift to MSA): الأردنيون طيبون جداً، لكن ضغوط الحياة جعلت الناس يعملون كثيراً ليعيشوا بكرامة.

The Gemini output is fluent, semantically accurate MSA — but fails the register criterion. SARI does not adequately penalize this because the MSA output may still share n-grams with the MSA-written gold reference. This is a task-compliance failure invisible to SARI.

Challenge 3: Dialectal Orthographic Inconsistency and SARI Bias Arabic dialects lack standardized written forms. Egyptian negation can be مش or مو; Moroccan Darija mixes French-origin words in Arabic script. SARI treats مش and ليس as different tokens even when they express identical negation.

This explains why Jais achieves high Dialect SARI (0.4435) despite register shift: its MSA output happens to share function words with both the source and the reference (since some gold references also use MSA-adjacent forms). The bias also explains why all models score lower on Dialect than MSA regardless of output quality.

Two Difficult or Borderline Cases

Borderline Case 1: Theological Fidelity vs. Compression (P021, Classical) P021 describes Caliph Umar ibn al-Khattab’s famine relief. The statement لئن مات جائع في الشام لأسألنَّ عنه يوم القيامة is historically and theologically significant. GPT-4o retained it; Jais (valid Classical output) dropped it entirely. SARI partially rewards Jais for the deletion (DELETE increases) while penalizing it for losing reference n-grams. Human evaluation is required for classical religious texts where fidelity is paramount.

Borderline Case 2: When a Good Simplification Scores Worse (P018, Classical) For P018 (Sahih Muslim Hadith), Fanar scores SARI = 0.72 while GPT-4o scores only 0.38. Qualitatively, GPT-4o produces the more genuinely simplified output (strips transmission chain, simplifies vocabulary). Fanar retains the full chain of transmission in words close to the human reference. The higher Fanar score arises purely from reference proximity — it is more similar to the reference, not more simplified. This demonstrates that SARI conflates “close to reference” with “well simplified.”

Limitations

1. **Jais performs dialect-to-MSA conversion, not dialect simplification.** While Jais’s 40 outputs are all valid (non-echo) and score well numerically, it does not preserve dialectal register. Its high Dialect SARI (0.4435) reflects MSA outputs that partially match MSA-adjacent gold references, not genuine dialectal simplification competence.
2. **ALLaM and Fanar expand rather than simplify.** Both models produce outputs longer than the source (ratios 1.086 and 1.105). This is a fundamental task compliance failure. Their KEEP scores are structurally inflated and DELETE scores depressed.
3. **SARI underestimates true semantic agreement.** SARI IAA (0.4620) is substantially lower than manual Content Unit Agreement (90% Strong). SARI must be supplemented with semantic metrics (e.g., BERTScore using AraBERT or CAMELBERT) or human evaluation.
4. **SARI cannot detect hallucinations.** Fanar’s addition of “watching television” (P027) and ALLaM’s occasional topic drift receive no specific penalty beyond indirect n-gram mismatch.
5. **SARI conflates reference proximity with quality.** As shown in P018, a model making minor stylistic changes can score higher than one making genuine aggressive simplification.
6. **Dialectal orthographic inconsistency biases SARI.** Dialectal Arabic lacks standard orthography. SARI treats مش and ليس as different tokens even when they express identical negation.

7. **Single inference run per model.** All models were run once. Temperature and sampling stochasticity may affect results, particularly for web GUI models.
8. **No third annotator for dispute resolution.** The four Moderate-agreement cases (P005, P007, P010, P019) would benefit from a third annotator’s ruling.

Conclusions, Comments, and Recommendations

Conclusions

1. **GPT-4o is the best-performing model overall** (SARI = 0.4725, best on MSA at 0.5309). It is the only model approaching the human IAA ceiling on any variety.
2. **Jais ranks 2nd overall (0.4233) when all 40 outputs are properly included**, and leads the Dialect category (0.4435). Its strategy of converting dialect to MSA produces high SARI but violates register preservation guidelines.
3. **Gemini 1.5 Pro is a consistent 3rd-place model** (0.4223) with competitive performance across all three varieties.
4. **Fanar and ALLaM excel on Classical Arabic** (0.4498 and 0.4296) but fail on the length criterion.
5. **Dialectal Arabic is the primary unsolved challenge**, with register preservation being the key dimension that SARI alone cannot capture.
6. **Text compression is a necessary but unguaranteed behavior.** GPT-4o, Gemini, and Jais compress naturally. ALLaM and Fanar must be explicitly prompted.
7. **SARI is a necessary but insufficient metric for Arabic simplification.**

Recommendations

- **Add explicit register-preservation constraint to prompts for Jais:** Specify “preserve the original Arabic dialect” (حافظ على اللهجة العربية الأصلية) to test whether dialect-to-MSA conversion is instruction-dependent.
- **Add explicit length constraint to prompts for ALLaM and Fanar:** Specify a target word count (e.g., “no more than 60% of the original word count”).
- **Add BERTScore evaluation:** Use AraBERT or CAMELBERT to complement SARI’s lexical overlap with embedding-based meaning similarity.
- **Include human evaluation for Dialectal samples:** Rate a 10-paragraph Dialectal subset for (a) fluency, (b) meaning preservation, and (c) register consistency on a 5-point Likert scale.
- **Add faithfulness annotation:** Native speaker annotation for content not present in the source (hallucination detection).
- **Use multi-reference SARI as default:** Consistently reduces penalization of valid paraphrases.
- **Expand Dialectal coverage:** Add Sudanese, Yemeni, and Palestinian Arabic in future iterations.

Selected Model Outputs — Qualitative Comparison

Table 42: All Model Outputs for P018 (Classical Hadith, Sahih Muslim)

Model	SARI	Output
Gold A1	—	روى الإمام مسلم عن أبي هريرة أن النبي قال: من فرّج هم مسلم في الدنيا، فرّج الله همّه يوم القيامة. ومن يسّر على شخص في ضائقة، يسّر الله له في الدنيا والآخرة.
Fanar	0.72	روى الإمام مسلم في صحيحه عن أبي هريرة رضوان الله عليه أن رسول الله صلى الله عليه وسلم قال: من يخفف عن مؤمن همّاً في الدنيا، سيهون الله عنه هموم يوم القيامة.
Gemini	0.62	قال الرسول ﷺ إن من ساعد مسلماً في مشكلة في الدنيا، ساعده الله في مشاكله يوم القيامة. ومن سهل على شخص مديون، سهل الله عليه في الدنيا والآخرة.
ALLaM	0.60	قال النبي صلى الله عليه وسلم: من فرج عن مؤمن كربة من كرب الدنيا، فرج الله عنه كربة من كرب يوم القيامة. ومن يسر على معسر، يسر الله عليه في الدنيا والآخرة.
Jais	0.54	مسلم روى عن أبي هريرة أن النبي صلى الله عليه وسلم قال: من نفّس عن مؤمن كربة من كرب الدنيا، نفّس الله عنه كربة من كرب يوم القيامة.
GPT-4o	0.38	من يخفف عن شخص ضيقاً في الدنيا، يخفف الله عنه في الآخرة. ومن يسهل على محتاج، يسهل الله عليه في الدنيا والآخرة.

Qualitative note on P018: Fanar scores highest (0.72) by retaining the transmission chain close to the reference, while GPT-4o scores lowest (0.38) despite producing the most genuinely compressed output. SARI conflates reference proximity with simplification quality here, making human evaluation essential for classical religious texts.